

Fusion draft: SST-30/59 Atomic masses cluster

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About this fusion draft

This is an automatically assembled fusion draft: multiple SST manuscripts concatenated as chapters. It is intended for navigation and editing workflow, not as a final merged publication without manual cleanup.

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Chapter 1

SST-59 Part 1 — Mass hydrodynamic topology

**Hydrodynamic Topology as an Effective Model for Inertial Mass:
A Self-Contained Derivation of a Topology-Modulated Mass
Functional**

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Abstract

We present a self-contained, hydrodynamically motivated effective model in which the invariant rest mass of a localized, topologically stable excitation is identified with the integral of a localized kinetic-energy density over its effective support volume.

The model assumes an incompressible, inviscid background medium (Euler–Kelvin regime) supporting persistent filamentary defects whose stability is protected by circulation conservation. The resulting rest-mass functional factorizes into: (i) a universal energy-density scale $u = \frac{1}{2}\rho_{\text{core}}\|\mathbf{v}_0\|^2$, (ii) a geometric volume proxy $V = \pi r_c^3 L_{\text{tot}}$, and (iii) a dimensionless kernel $\mathcal{K}(T)$ depending only on a small set of topological indices. We motivate a minimal closed kernel of the form

$$M(T) = \mathcal{K}(T) \frac{u V(T)}{c^2}, \quad \mathcal{K}(T) = \left(\frac{4}{\alpha}\right)^{\mathcal{G}(T)} k(T)^{-3/2} \varphi^{-g(T)} n(T)^{-1/\varphi}, \quad (1.1)$$

where k is a complexity index (e.g. braid index/crossing proxy), g is Seifert genus, n is the number of components, α is the fine-structure constant, and φ is the golden ratio. The paper is written to be “orthodox-friendly”: all strong statements are framed as model assumptions, the status of each scaling law is clearly labeled, and falsifiable preferred-frame and spectrum-selection consequences are listed. We also separate background density ρ_f (governing wave transport) from a core-localized density ρ_{core} (governing inertial mass) and explain why this separation is required for dimensional consistency and empirical scale matching.

Scope & status. This paper (Part I) constructs a mathematically controlled *effective* model in which invariant rest mass is represented by a localized energy functional of a persistent filamentary excitation in the Euler–Kelvin regime. No microscopic ontology is assumed beyond the field-theoretic structure stated below.

Assumptions (Euler–Kelvin regime).

- Incompressible flow: $\nabla \cdot \mathbf{v} = 0$.
- Inviscid dynamics: Euler limit (no viscosity, no dissipation).
- Barotropic closure: $p = p(\rho)$ so that Kelvin circulation conservation applies.
- No reconnection / topology preserving evolution: filament topology is conserved over the timescales of interest.
- Persistent localized excitation: the energy density is spatially localized (core + envelope) and admits a time-averaged stationary description.

What is claimed vs. not claimed.

- Claimed: In this regime one can define a rest-energy functional E_0 for a localized filamentary excitation such that $M \equiv E_0/c^2$ behaves operationally as an inertial mass parameter (in the sense of low-velocity response and resonance scaling).
- Not claimed: We do not claim that “matter is a fluid” or that any specific microscopic substrate is established; the construction is presented strictly as an effective Hamiltonian functional model with well-defined assumptions.

Observable definition (what is measured). The primary observable is the *invariant rest-energy functional* of a localized excitation,

$$E_0 \equiv \int_V \varepsilon(\mathbf{x}) dV, \quad M \equiv \frac{E_0}{c^2}, \quad (1.2)$$

where $\varepsilon(\mathbf{x})$ is the localized (time-averaged) energy density associated with the persistent filamentary configuration.

Operationally, M is the parameter that enters low-velocity inertial response and mode scaling; e.g. for any probe whose eigenfrequency depends on the effective inertia, one tests the scaling $f \propto M^{-1/2}$ (holding geometry and stiffness fixed), thereby linking the functional prediction E_0 to measurable frequency shifts or acceleration response.

1.1 Introduction and scope

Assumptions (Euler–Kelvin regime). Throughout, we work in the incompressible, inviscid Euler limit and restrict attention to the Kelvin-admissible sector in which circulation is materially conserved. Concretely:

- **Incompressible flow:** $\nabla \cdot \mathbf{v} = 0$.
- **Inviscid Euler dynamics:** $\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -(1/\rho) \nabla p$ (no viscosity).
- **Barotropic closure:** $p = p(\rho)$ (needed for Kelvin circulation conservation).
- **Circulation conservation (Kelvin theorem):** the circulation $\Gamma = \oint_{\mathcal{C}(t)} \mathbf{v} \cdot d\boldsymbol{\ell}$ is invariant for material loops $\mathcal{C}(t)$.
- **No reconnection / fixed topology during evaluation:** filament/link topology is treated as invariant over the time window of interest (i.e. the configuration remains in a fixed topological sector).

These assumptions define the controlled regime in which persistent filamentary excitations are well-posed objects for an effective mass functional. (The “Euler–Kelvin + no reconnection” set is explicit as a condition in the Rosetta card [?].)

What is claimed vs. what is not claimed.

- **Claimed:** (i) invariant rest mass can be represented operationally as $M = E_{\text{rest}}/c^2$ with E_{rest} given by a localized energy functional for a persistent filamentary excitation; (ii) the proposed construction admits a clean factorization into a scale $\times$ a geometric volume proxy $\times$ a dimensionless topology-dependent kernel, enabling falsifiable downstream mappings.
- **Not claimed:** (i) no commitment is made to a unique microscopic substrate or ontology; the construction is framed as a mathematically controlled effective model; (ii) specific kernel choices (e.g. exponents or discrete suppression factors) are not asserted as theorems unless explicitly proven—they remain testable modeling hypotheses.

Observable definition (what is actually measured). The primary observable is the *rest mass* inferred from rest-energy via

$$M \equiv \frac{1}{c^2} \int_V \rho_E(\mathbf{x}) dV,$$

with ρ_E identified with the organized, localized energy density of the excitation core (kinetic plus any explicitly retained topological/pressure contribution). The operational comparison target is standard: M is the parameter that governs inertial response, e.g. $F = Ma$ for center-of-mass motion in the nonrelativistic limit, and $E_{\text{rest}} = Mc^2$ when comparing against tabulated rest energies. The theory’s output is therefore a concrete map

$$(\text{configuration class}) \longmapsto M$$

that can be benchmarked against measured masses and (in later parts) against kinematically induced corrections.

The microscopic origin of inertial mass remains an open question spanning particle physics, condensed matter analogies, and quantum field theory. In the Standard Model (SM), fermion masses arise from Yukawa couplings to the Higgs condensate, with parameters not predicted by the theory [5, 6, 7]. This motivates exploration of alternative *effective* mechanisms in which inertia emerges from localized energy storage and topological stability.

This article develops a hydrodynamic-topology effective model where: (i) persistent localized excitations are protected by circulation conservation (Kelvin theorem), (ii) their rest mass equals a localized kinetic-energy content divided by c^2 , (iii) topology modulates the efficiency of energy localization through a compact kernel $\mathcal{K}(T)$.

Status note. Equation (3.19) is presented as a *closed effective ansatz* motivated by Euler–Kelvin hydrodynamics, slender-filament energetics, and discrete scale invariance considerations. Certain exponent choices (notably $-3/2$) admit supporting scaling arguments; others (golden-ratio exponents) are principled but not unique. The goal is a journal-ready, checkable formulation with explicit assumptions and falsifiers.

1.2 Hydrodynamic substrate and topological stability

1.2.1 Euler–Kelvin regime

Assume a background medium described (on relevant scales) as incompressible and inviscid:

$$\nabla \cdot \mathbf{v} = 0, \quad \partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p, \quad (1.3)$$

with barotropic closure $p = p(\rho)$ locally. In this regime, Kelvin’s circulation theorem implies that the circulation along a material loop is conserved:

$$\Gamma = \oint_{\mathcal{C}(t)} \mathbf{v} \cdot d\boldsymbol{\ell} = \text{const}. \quad (1.4)$$

Consequently, filamentary circulation structures cannot smoothly terminate; closed loops are dynamically preferred and topologically robust [8, 6, 7].

1.2.2 Filamentary defects as persistent excitations

We model a localized excitation T as a closed filament (or link of n components) with a microscopic core radius r_c and a characteristic tangential swirl speed $\|\mathbf{v}_\mathcal{O}\| = \|\mathbf{v}_\mathcal{O}\|$. The physical picture is a slender tube of radius r_c carrying a quantized circulation scale Γ_0 , which we parameterize as

$$\Gamma_0 \equiv 2\pi r_c \|\mathbf{v}_\mathcal{O}\|. \quad (1.5)$$

Equation (1.5) is a definition of the characteristic scale; quantization of Γ is treated as an additional postulate compatible with quantized vortices in superfluids [11, 12].

1.3 Mass as localized kinetic energy

1.3.1 Covariant definition (orthodox framing)

In relativistic field theory, invariant rest mass is the rest-frame energy divided by c^2 . We adopt the operational definition: in the excitation rest frame,

$$M \equiv \frac{1}{c^2} \int_V \rho_E(\mathbf{x}) dV, \quad (1.6)$$

where ρ_E is the localized energy density attributed to the excitation and V encloses its support. This is a standard bookkeeping identity once ρ_E is defined.

1.3.2 Core-localized energy density scale

Assume the excitation has a core region characterized by an effective core density ρ_{core} (distinct from the background density ρ_f that governs long-wavelength wave transport). The simplest kinetic-energy density scale is then

$$u \equiv \frac{1}{2} \rho_{\text{core}} \|\mathbf{v}_0\|^2, \quad (1.7)$$

with units J m^{-3} . We emphasize: (A.8) is the *local* kinetic-energy density of the organized core motion, not the background vacuum energy.

Why separate ρ_f and ρ_{core} ? If one attempted to build particle-scale masses by integrating background density ρ_f over microscopic volumes, the resulting scale is generically far too small. The effective model requires a localized “core phase” with much larger inertial density to achieve observed masses while keeping the background “light.” This is analogous in spirit (not identity) to the separation between condensate order-parameter stiffness and quasiparticle effective masses in superfluid/superconducting systems.

1.4 Geometry: volume proxy from ropelength

1.4.1 Slender-filament volume approximation

For a slender tube of radius r_c and centerline length ℓ , the volume is approximately

$$V \simeq \pi r_c^2 \ell. \quad (1.8)$$

Write $\ell = r_c L_{\text{tot}}$, where L_{tot} is a dimensionless ropelength-like quantity (e.g. length in units of r_c). Then

$$V(T) = \pi r_c^3 L_{\text{tot}}(T). \quad (1.9)$$

The use of a minimal (tight) ropelength as a ground-state proxy is consistent with the idea that line tension drives contraction until self-avoidance/self-interaction prevents further shrinkage [7].

1.5 Topology: a minimal closed kernel

1.5.1 Factorized master formula

Combining (1.6), (A.8), and (A.9), we obtain the factorized form

$$M(T) = \mathcal{K}(T) \frac{u V(T)}{c^2} = \mathcal{K}(T) \frac{\left(\frac{1}{2} \rho_{\text{core}} \|\mathbf{v}_0\|^2\right) (\pi r_c^3 L_{\text{tot}}(T))}{c^2}. \quad (1.10)$$

All nontrivial dependence on topology and “energy-localization efficiency” is now placed into $\mathcal{K}(T)$.

1.5.2 Minimal kernel choice and interpretation

We consider the dimensionless kernel

$$\mathcal{K}(T) = \left(\frac{4}{\alpha}\right)^{\mathcal{G}(T)} k(T)^{-3/2} \varphi^{-g(T)} n(T)^{-1/\varphi}. \quad (1.11)$$

Here:

- $k(T)$ is a complexity index (e.g. braid index, or a crossing-number proxy) representing coarse geometric/topological complexity relevant for self-induction and screening.
- $g(T)$ is Seifert genus, capturing surface complexity of a spanning surface.
- $n(T)$ is number of components (for links).
- $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio, used as a discrete-scale modulation factor (see below).
- α is the electromagnetic fine-structure constant, used here as a universal dimensionless normalization connecting energy localization to a measured coupling strength; we treat this as a model postulate rather than a theorem.

Status of each factor.

1. The factor $(4/\alpha)^{G(T)}$ is a *gated normalization postulate*. When $G(T) = 0$, the excitation is in a “shielded” sector and the vacuum-coupling amplification is inactive; when $G(T) = 1$, the amplification is active.
2. The exponent $-3/2$ for $k(T)$ admits scaling support from diffusion/heat-kernel analogies and from dimensional considerations in filament energetics; nevertheless, it remains an effective choice.
3. The φ^{-g} and $n^{-1/\varphi}$ factors encode *discrete scale invariance (DSI)*-motivated suppression with increasing topological obstruction and multiplicity. These are principled but not unique; alternative monotone suppressions can be tested.

1.5.3 Formal shielding gate via rational sliceness (hard constraint)

The narrative “shielding vs. exposure” is promoted here to a *checkable gate* $G(T) \in \{0, 1\}$ defined using rational sliceness (rational homology 4-ball bounding). This provides an explicit on/off rule for when the factor $4/\alpha$ is active.

Definition 1.5.1 (Rational homology 4-ball). A smooth compact 4-manifold W is a *rational homology 4-ball* if $\partial W \cong S^3$ and

$$H_*(W; \mathbb{Q}) \cong H_*(B^4; \mathbb{Q}).$$

Definition 1.5.2 (Rational sliceness). A knot $K \subset S^3$ is *rationally slice* if there exists a rational homology 4-ball W with $\partial W \cong S^3$ in which K bounds a smoothly embedded disk $D^2 \subset W$. Equivalently, K is slice in some W that is “ B^4 up to $\mathbb{Q}$ -homology.” [5]

Definition 1.5.3 (Shielding gate). Define the *shielding-gate exponent* $G(T) \in \{0, 1\}$ by

$$G(T) = \begin{cases} 0, & \text{if } T \text{ is rationally slice (shielded sector),} \\ 1, & \text{otherwise (exposed sector).} \end{cases}$$

In particular, $(4/\alpha)^{G(T)} = 1$ in the shielded sector and equals $4/\alpha$ in the exposed sector.

Double-twist family as an explicit, computable gate. For the double twist knots $K_{m,n}$ (two twist regions with integer parameters m, n), rational sliceness is completely characterized. We use this as a *hard constraint* (not merely inspiration) because it yields an exact, finite test for $G(K_{m,n})$.

Theorem 1.5.1 (Rational sliceness of double twist knots [5]). *A double twist knot $K_{m,n}$ is rationally slice if and only if*

$$mn = 0 \quad \text{or} \quad n = -m \quad \text{or} \quad n = -m \pm 1.$$

Remark 1.5.1 (Gate specialization for $K_{m,n}$). The gate becomes the explicit indicator of *not* satisfying the above condition:

$$G(K_{m,n}) = \begin{cases} 0, & mn = 0 \text{ or } n = -m \text{ or } n = -m \pm 1, \\ 1, & \text{otherwise.} \end{cases}$$

The proof in [5] uses definite 4-manifold obstructions (Donaldson-type input) to rule out all other cases. [18]

1.5.4 DSI motivation (orthodox phrasing)

Discrete scale invariance appears in multiple contexts (fracture, turbulence cascades, renormalization-group limit cycles) and can generate preferred ratios in spectra [15]. In the present model, we adopt DSI as an organizing principle: increased topological obstruction (g) and component count (n) reduces the fraction of volume that can maintain coherent core motion, implemented via the golden-ratio suppression in (A.10). This is a hypothesis, testable by spectrum-fitting stability and by predictive out-of-sample checks.

1.6 Representative selection vs. mass kernel

Equation (1.10) depends on $L_{\text{tot}}(T)$, which in practice requires selecting a representative configuration within a topological sector. To keep the mass kernel (A.10) minimal, we separate:

1. **Selection functional:** choose a representative K^* within a sector by minimizing an effective topological energy

$$E_{\text{eff}}[K] = \alpha_C C(K) + \beta_L L(K) + \gamma_H H(K), \quad K^* = \arg \min E_{\text{eff}}[K]. \quad (1.12)$$

Here C, L, H are monotone complexity/stability proxies; $\alpha_C, \beta_L, \gamma_H$ are weights. This stage defines which “tight” representative is used to compute L_{tot} and indices.

2. **Mass evaluation:** compute $(k, g, n, L_{\text{tot}})$ on K^* and evaluate (3.19).

This separation prevents double-counting complexity inside $\mathcal{K}$ and keeps the mass map compact.

1.7 Calibration, prediction program, and limitations

1.7.1 Calibration boundary

Any model with a universal scale must be anchored. A minimal approach is to calibrate one reference mass (e.g. the electron) by choosing L_{tot} of its representative so that (3.19) matches M_e . All other masses become predictions conditional on the selected topological assignments and computed indices/ropelengths.

1.7.2 Composite systems and binding energy

For nuclei and atoms, a first-pass assembly model is

$$M_{\text{atom}}(Z, N) \approx Z M_p + N M_n + Z M_e - \frac{E_{\text{bind}}(Z, N)}{c^2}, \quad (1.13)$$

where each constituent mass (e.g. M_p, M_n, M_e) is supplied by the same master kernel $M(T)$ evaluated on the chosen topological representatives, while E_{bind} is provided by a *bridge model* until a fully hydrodynamic binding theory is derived.

SEMF bridge model (drop-in compatible with the master kernel)

As an orthodox-accurate bridge, take the Bethe–Weizsäcker semi-empirical mass formula (SEMF) binding energy:

$$E_{\text{bind}}^{\text{SEMF}}(Z, N) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} + \delta(A, Z), \quad A = Z + N, \quad (1.14)$$

with the pairing term

$$\delta(A, Z) = \begin{cases} +a_P A^{-1/2}, & Z, N \text{ even,} \\ 0, & A \text{ odd,} \\ -a_P A^{-1/2}, & Z, N \text{ odd.} \end{cases} \quad (1.15)$$

This keeps the atomic/nuclear *bookkeeping* orthodox while delegating the *constituent rest masses* to the topology-modulated kernel (3.19). [8, 14]

Optional SST-compatible modulation (bridge extension)

If one wishes to encode “topological packing efficiency” at the nuclear scale without changing the SEMF structure, promote (a_V, a_S) to mild A -dependent functions through a monotone coherence factor $\Xi(A)$ (fit/validated on data):

$$a_V(A) = a_V^{(0)} \Xi(A), \quad a_S(A) = a_S^{(0)} \Xi(A)^{-1/2}, \quad (1.16)$$

leaving a_C, a_A, a_P as orthodox baselines at first pass. This yields an SST-SEMF bridge that remains algebraically identical to (1.14) but allows controlled deviations for large A in out-of-sample tests.

1.7.3 Preferred-frame (foliation) extension and falsifiers

If the background medium defines a preferred rest frame, motion relative to it can generate small preferred-frame corrections. Parametrize the effective mass for an excitation moving at velocity v relative to the medium by

$$M_{\text{eff}}(v) = M_0 \left[1 + \eta_1 \frac{\mathbf{v} \cdot \hat{\mathbf{u}}}{c} + \eta_2 \frac{(\mathbf{v} \cdot \hat{\mathbf{u}})^2}{c^2} + \dots \right], \quad (1.17)$$

where $\hat{\mathbf{u}}$ is the medium rest-frame direction field and η_i map to preferred-frame phenomenology constrained by precision tests (PPN/SME literature) [20, 21]. The model is falsified if required η_i violate existing bounds or if predicted anisotropies are not observed.

1.8 Numerical sanity check (dimension and scale)

Define the universal prefactor

$$M_0 \equiv \frac{u \pi r_c^3}{c^2} = \frac{\left(\frac{1}{2} \rho_{\text{core}} \|\mathbf{v}_{\text{O}}\|^2\right) \pi r_c^3}{c^2}, \quad M_1 \equiv \left(\frac{4}{\alpha}\right) M_0. \quad (1.18)$$

Then

$$M(T) = M_0 \left(\frac{4}{\alpha}\right)^{\mathcal{G}(T)} \left[k(T)^{-3/2} \varphi^{-g(T)} n(T)^{-1/\varphi} \right] L_{\text{tot}}(T). \quad (1.19)$$

Equation (1.19) makes explicit that the model requires only a universal base mass scale M_0 , a *gated* amplification $(4/\alpha)^{\mathcal{G}(T)}$, and dimensionless topology/geometry factors.

1.9 Resolution Consistency and Gradient Expansion of the Mass Functional

1.9.1 Motivation

The invariant mass functional derived in Sec. ?? was expressed in terms of the swirl energy density

$$\rho_E^{(0)} = \frac{1}{2} \rho_f \|\mathbf{v}_\Omega\|^2. \quad (1.20)$$

However, any quadratic nonlinearity subjected to spatial coarse-graining generates subgrid contributions that cannot be captured solely by the zeroth-order (local) kinetic term. This is a general consequence of filtering theory and the multi-scale gradient expansion of quadratic stresses [1, 2, 3, 4].

Therefore, to ensure resolution-consistency of the mass functional, we include the leading curvature correction induced by coarse-graining.

1.9.2 Gradient Expansion under Coarse-Graining

Let G_ℓ denote a symmetric filter of width ℓ . Define the coarse-grained velocity field

$$\bar{\mathbf{v}} = G_\ell * \mathbf{v}. \quad (1.21)$$

The filtered quadratic mismatch (subgrid stress analogue) is

$$\tau_{ij} = \overline{v_i v_j} - \bar{v}_i \bar{v}_j. \quad (1.22)$$

For smooth fields and symmetric filters, τ_{ij} admits an even-order gradient expansion:

$$\tau_{ij} = \kappa_2 \ell^2 \partial_k \bar{v}_i \partial_k \bar{v}_j + \kappa_4 \ell^4 \partial_{kl} \bar{v}_i \partial_{kl} \bar{v}_j + O(\ell^6), \quad (1.23)$$

where κ_2, κ_4 are dimensionless constants determined by the filter kernel.

This hierarchy is mathematically unavoidable for filtered quadratic nonlinearities [3].

1.9.3 Effective Swirl Energy Density

The leading resolution-consistent local truncation of the swirl energy density is therefore

$$\rho_E^{\text{eff}}(\ell) = \frac{1}{2} \rho_f \left(\|\bar{\mathbf{v}}\|^2 + \kappa_2 \ell^2 \|\nabla \bar{\mathbf{v}}\|^2 \right), \quad (1.24)$$

with

$$\|\nabla \bar{\mathbf{v}}\|^2 = \partial_i \bar{v}_j \partial_i \bar{v}_j. \quad (1.25)$$

Dimensional consistency is immediate:

$$[\rho_f] = \text{kg m}^{-3}, \quad [\ell^2 \|\nabla \bar{\mathbf{v}}\|^2] = \text{m}^2 \text{s}^{-2}, \quad [\rho_E] = \text{J m}^{-3}.$$

Higher-order curvature terms $O(\ell^4)$ may be included, but Eq. (1.24) represents the minimal local truncation.

1.9.4 Resolution-Consistent Mass Functional

The coarse-grained invariant mass functional becomes

$$M(\ell) = \frac{1}{c^2} \int \rho_E^{\text{eff}}(\ell) dV. \quad (1.26)$$

Explicitly,

$$M(\ell) = \frac{\rho_f}{2c^2} \int \left(\|\bar{\mathbf{v}}\|^2 + \kappa_2 \ell^2 \|\nabla \bar{\mathbf{v}}\|^2 \right) dV. \quad (1.27)$$

Let L denote the characteristic spatial variation scale of the localized swirl structure. Then

$$\frac{\Delta M_2}{M_0} \sim \kappa_2 \left(\frac{\ell}{L} \right)^2. \quad (1.28)$$

Hence:

- If $L \gg \ell$, curvature corrections are negligible.
- If $L \sim \ell$, corrections become $O(1)$.

Therefore, resolution-consistent mass evaluation for localized swirl structures requires inclusion of the gradient term.

1.9.5 Falsifiable Diagnostic

For a fixed physical configuration, compute $M(\ell)$ over a sequence of filter scales $\ell = n \delta x$.

Resolution consistency requires

$$\text{Var}_\ell(M(\ell)) < \text{Var}_\ell(M_0(\ell)). \quad (1.29)$$

Failure of this condition indicates either:

- insufficient derivative accuracy,
- breakdown of locality,
- or necessity of nonlocal kernel corrections.

This diagnostic provides a direct computational test of the resolution-consistent mass hypothesis.

10-year-old analogy (one sentence). A particle is modeled like a tiny knotted loop of spinning fluid: its mass is how much “spinning energy” is packed into the loop, and complicated knots pack that energy less efficiently.

1.10 Conclusions

We have presented a self-contained hydrodynamic-topology effective model for inertial mass in which rest mass equals localized kinetic energy divided by c^2 , and topology enters through a minimal dimensionless kernel multiplying a filament volume proxy. The framework is explicitly modular: (i) covariant mass functional, (ii) core energy density scale, (iii) slender-filament geometry, (iv) topology-modulated localization efficiency, and (v) optional representative selection by an energy functional. The approach yields clear falsifiers (preferred-frame constraints, out-of-sample mass predictions, stability of selection under kernel variations) and provides a compact, checkable formula suitable for systematic benchmark testing.

Acknowledgments

None.

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Chapter 2

SST-59 Part 2 — Mass generation / topological shielding

Rational-Concordance Constraints (Double-Twist) and an SEMF Bridge to Atomic Masses

Abstract

We present a self-contained effective mass model in which invariant rest mass is identified with a localized swirl-energy integral and modulated by topology through a compact, testable kernel. The central deliverable is a single factorized master equation

$$M(T) = \left(\frac{4}{\alpha}\right)^{S(T)} k(T)^{-3/2} \phi^{-g_4(T)} n(T)^{-1/\phi} \left(M_0 L_{\text{ideal}}(T)\right),$$

where L_{ideal} is a dimensionless ropelength proxy, k is a spectral/complexity index, n is component count, and g_4 is the *slice genus* (a concordance-invariant genus measure). We explicitly constrain the kernel by requiring robustness under (rational) concordance moves and, as a concrete genus-one testbed, we incorporate the double-twist family $K_{m,n}$ and its slice criterion, which provides a sharp on/off benchmark for the genus factor [5]. Non-original ingredients (ropelength energetics, discrete scale invariance, concordance constraints) are cited; SST-specific identifications are labeled as postulates.

New in SST-59 (hard-formalized): (i) we replace the narrative “shielding” discussion by a *binary shielding gate* $G(T) \in \{0, 1\}$ that decides whether the amplification $(4/\alpha)$ is active; (ii) we incorporate rational concordance of double-twist knots $K_{m,n}$ as a *hard constraint* on G via a theorem-level characterization of rational sliceness; and (iii) we add a bridge model to atomic masses by subtracting an SEMF-like binding functional that is explicitly compatible with the master kernel.

Scope & status. This paper (Part II) formalizes *topological admissibility constraints* on topology-indexed mass kernels used in the effective mass functional program. The goal is not to postulate new forces but to restrict the space of allowed kernel assignments by computable 4D-topological criteria.

Assumptions (Euler–Kelvin + topological sectoring).

- **Same dynamical regime as Part I:** incompressible, inviscid, barotropic; no reconnection.
- **Topological sector decomposition:** configurations are partitioned into equivalence classes T (knot/link types plus admissible deformations) with well-defined invariants.
- **Kernel ansatz:** a dimensionless kernel $\mathcal{K}(T)$ is permitted only when T satisfies a stated admissibility criterion; otherwise it is excluded from the model class.
- **No continuous Yukawa-like tuning:** only discrete indices/invariants label admissible kernel structure in this part.

What is claimed vs. not claimed.

- **Claimed:** A binary admissibility map $G(T) \in \{0, 1\}$ can be defined from concordance-based (4D-topological) constraints and used as a *selection rule* that restricts the allowed kernel classes $\mathcal{K}(T)$.
- **Not claimed:** We do not claim a proven microscopic necessity that a specific constant (e.g. $4/\alpha$) is “dictated” by a single invariant; rather, we present a falsifiable constraint program: certain kernel normalizations/ansätze are allowed or excluded by computable topological criteria.

Observable definition (what is measured). The observable output of Part II is a set of *testable admissibility predictions*: given a configuration class T , the model predicts either

$$G(T) = 1 \text{ (admissible)} \quad \text{or} \quad G(T) = 0 \text{ (excluded)}, \quad (2.1)$$

together with the induced restriction on the mass-kernel assignment $\mathcal{K}(T)$. These predictions become measurable in Part III by checking whether only admissible classes can consistently reproduce mass hierarchies under a single fixed normalization, and by identifying outliers as violations of the admissibility criterion rather than as fit freedom.

2.1 Introduction

Standard Model particle masses are currently free parameters determined by Yukawa couplings to the Higgs field. Swirl String Theory (SST) proposes an alternative origin: mass is the invariant energy of a topological defect (vortex) in a continuous medium.

In this framework, the rest mass M is defined by the volume integral of the swirl energy density ρ_E :

$$M = \frac{1}{c^2} \iiint_V \left(\frac{1}{2} \rho_{\text{core}} \|\mathbf{v}_\text{C}\|^2 + P_{\text{top}} \right) dV \quad (2.2)$$

This paper derives the scaling laws that map knot topology (T) to physical mass, identifying three key mechanisms: geometric ropelength, spectral suppression, and vacuum coupling amplification.

2.2 The SST Master Equation

We propose the following unified mass formula for a stable topological defect T :

$$M(T) = \underbrace{\left(\frac{4}{\alpha}\right)^{G(T)}}_{\text{Vacuum Coupling (gate)}} \cdot \underbrace{n^{-1/\phi}}_{\text{Coherence}} \cdot \underbrace{k^{-3/2}}_{\text{Spectral}} \cdot \underbrace{(M_0 \cdot L_{\text{ideal}}(T))}_{\text{Geometric Base}} \quad (2.3)$$

Where:

- M_0 : The fundamental vortex mass unit per unit dimensionless length.
- L_{ideal} : The ropelength of the ideal knot configuration.
- k : The dominant spectral mode (symmetry order) of the knot.
- n : The number of linked components.
- α^{-1} : The fine-structure constant (≈ 137.036).
- $G(T) \in \{0, 1\}$: the *shielding gate*. $G(T) = 0$ means “perfect shielding” and $(4/\alpha)$ is *off*; $G(T) = 1$ means shielding is broken and $(4/\alpha)$ is *on*. Section 2.3 makes this formal.
- ϕ : The Golden Ratio (≈ 1.618), governing topological coherence.

2.3 A formal shielding gate from rational sliceness

2.3.1 Why a gate (not a narrative)

The master kernel uses a large discrete amplification scale $(4/\alpha)$. To keep the model checkable, the activation of this factor must be *decidable* from topology/invariants rather than prose. We therefore define a binary gate $G(T)$.

2.3.2 Rational concordance background (definitions)

Convention. We work in the smooth category.

Definition 2.3.1 (Rationally slice knot). A knot $K \subset S^3$ is *rationally slice* if it bounds a smoothly embedded disk $D \subset W$ in a smooth 4-manifold W with $\partial W = S^3$ and $H_*(W; \mathbb{Q}) \cong H_*(B^4; \mathbb{Q})$.

Definition 2.3.2 (Rational concordance group $\mathcal{C}_{\mathbb{Q}}$). Two knots $K_0, K_1 \subset S^3$ are *rationally concordant* if $K_0 \# -K_1$ is rationally slice. The set of equivalence classes under connected sum forms an abelian group $\mathcal{C}_{\mathbb{Q}}$ with identity the class of rationally slice knots.

2.3.3 Shielding gate (axiomatically tied to $\mathcal{C}_{\mathbb{Q}}$)

Definition 2.3.3 (Shielding gate). Define the shielding gate

$$G(T) = \begin{cases} 0, & \text{if the knot/link type assigned to } T \text{ is rationally slice (i.e. } [T] = 0 \text{ in } \mathcal{C}_{\mathbb{Q}}) \\ 1, & \text{otherwise.} \end{cases}$$

Equivalently, G turns $(4/\alpha)$ *off* exactly on the rationally-slice (“perfectly shielded”) topological sector.

Operational consequence. Given $G(T)$, the mass kernel is now strictly:

$$M(T) = \left(\frac{4}{\alpha}\right)^{G(T)} n^{-1/\phi} k^{-3/2} M_0 L_{\text{ideal}}(T).$$

2.4 Double-twist knots as a hard constraint on the gate

2.4.1 Definition of the family $K_{m,n}$

Definition 2.4.1 (Double twist knot $K_{m,n}$). Let $K_{m,n}$ denote the double-twist knot with integer parameters m, n , defined by the standard two-box double-twist diagram with m full twists in one twist region and n full twists in the other (sign convention as in [5]).

2.4.2 Exact rational sliceness characterization

Theorem 2.4.1 (Rational sliceness of $K_{m,n}$). *The double twist knot $K_{m,n}$ is rationally slice if and only if*

$$mn = 0 \quad \text{or} \quad n = -m \quad \text{or} \quad n = -m \pm 1.$$

Source: Theorem 1.1 of [5]. Earlier work established the “if” direction for subfamilies (e.g. $K_{m,-m}$ rationally slice); the converse is the hard content used here as a constraint.

[Explicit shielding gate on $K_{m,n}$] For the double-twist family,

$$G(K_{m,n}) = \begin{cases} 0, & mn = 0 \text{ or } |m + n| \leq 1, \\ 1, & \text{otherwise.} \end{cases}$$

Model use. Whenever a particle assignment uses a double-twist sector $K_{m,n}$, G is no longer adjustable: it is fixed by Corollary 2.4.2. This converts “shielding” into a falsifiable topological decision rule.

2.5 Bridge to atomic masses via an SEMF-compatible binding functional

2.5.1 What SEMF is (and why it is a bridge)

The semi-empirical mass formula (SEMF, Bethe–Weizsäcker liquid-drop model) approximates nuclear binding energy using volume/surface/Coulomb/asymmetry/pairing terms. In SST-59 it functions as a *phenomenological bridge*: it corrects the sum of constituent SST rest masses by a binding-energy subtraction, without modifying the fundamental particle kernel.

2.5.2 Atomic mass master equation (kernel + binding subtraction)

Let $A = Z + N$. Define the atomic mass (isotope) as

$$M_{\text{atom}}(Z, N) = Z M_p^{\text{SST}} + N M_n^{\text{SST}} + Z M_e^{\text{SST}} - \frac{E_{\text{bind}}^{\text{bridge}}(Z, N)}{c^2}, \quad (2.4)$$

where each $M_{p,n,e}^{\text{SST}}$ is computed from the SST-59 kernel, and $E_{\text{bind}}^{\text{bridge}}$ is the bridge functional.

2.5.3 SEMF bridge functional (drop-in)

We take

$$E_{\text{bind}}^{\text{bridge}}(Z, N) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} + \delta(A, Z), \quad (2.5)$$

with pairing

$$\delta(A, Z) = \begin{cases} +a_P A^{-1/2}, & Z, N \text{ even,} \\ 0, & A \text{ odd,} \\ -a_P A^{-1/2}, & Z, N \text{ odd.} \end{cases}$$

Equation (2.5) is standard (orthodox) nuclear phenomenology; within SST-59 it is explicitly a *bridge model*, not a derivation. Any SST-specific topological modulation (optional) must enter as a controlled deformation of the coefficients a_i , leaving the structure intact.

2.6 Derivation of Factors

2.6.1 Geometric Base Mass (M_0)

Using the canonical SST constants for the vortex core radius r_c , core density ρ_{core} , and swirl velocity $v_\odot$, the unit mass is derived as:

$$M_0 = \frac{\pi r_c^3 \cdot \frac{1}{2} \rho_{\text{core}} v_\odot^2}{c^2} \approx 2.277 \times 10^{-31} \text{ kg} \approx \mathbf{0.1277 \text{ MeV}/c^2} \quad (2.6)$$

This defines the energy scale of a "naked" vortex filament of length $L = r_c$.

2.6.2 Spectral Suppression ($k^{-3/2}$)

Treating the vortex loop as a dynamical object subject to Beltrami flow conditions ($\nabla \times \mathbf{u} = \lambda \mathbf{u}$), the partition function Z for quantum fluctuations is dominated by the 1-loop determinant. For a mode k with eigenvalue $\lambda_k \propto k$ in 3 spatial dimensions:

$$Z_k \propto (\det \mathcal{O}_k)^{-1/2} \propto (k^3)^{-1/2} = k^{-3/2} \quad (2.7)$$

This factor suppresses the effective mass of high-frequency (complex) knot modes.

2.7 Results and Validation

Particle	Topology	Geometry (M_{geom})	Factors	SST Mass	Exp. Mass
Electron	3_1 (Trefoil)	2.09 MeV	$k = 3, S = 0$	0.40 MeV	0.511 MeV
Muon	5_1 (Cinquefoil)	3.38 MeV	$k = 5, S \approx 0.38$	95.6 MeV	105.7 MeV
Proton	$3 \times (5_2/6_1)$	$\sim 10.2 \text{ MeV}$	$n = 3, S = 1$	939.8 MeV	938.3 MeV

Table 2.1: Comparison of predicted masses versus experimental values.

2.7.1 The Proton-Electron Ratio

The model successfully bridges the gap between the electron scale ($\sim 0.5 \text{ MeV}$) and the nucleon scale ($\sim 1 \text{ GeV}$). The proton mass is generated not by geometric length alone, but by the *amplification* of the geometric mass via the vacuum coupling factor ($4/\alpha$) and the coherence factor for $n = 3$ components ($3^{-1/\phi} \approx 0.5$).

2.8 Conclusion

Swirl String Theory provides a geometric basis for the mass hierarchy. The distinction between generations is identified as a topological phase transition in vacuum shielding. This framework eliminates the need for arbitrary Yukawa couplings, replacing them with discrete topological invariants (k, n, g) and a single fundamental mass scale M_0 .

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Chapter 3

SST-59 Part 3 — Atomic masses from topological invariants

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Abstract

We present a classical, topology-driven mass formula for elementary particles, atomic nuclei, and molecules. The theory is implemented numerically in a companion reference code and is derived here in a form exactly isomorphic to that implementation. Mass emerges as a geometric invariant associated with knotted field configurations, independent of quantum postulates or Higgs couplings. The theory depends only on geometric and topological quantities: a kernel suppression index, genus, component number, and dimensionless ropelength.

Scope & status. This paper (Part III) provides a reproducible benchmark study of a topology-indexed mass functional across a curated dataset of nuclei/atoms/molecules. It separates (i) constituent mass generation via a dimensionless topological kernel from (ii) binding-energy phenomenology via a documented bridge model.

Assumptions (benchmark semantics).

- **Kernel discreteness:** constituent masses are generated from discrete topology indices (no per-element continuous fitting).
- **Single normalization:** the absolute mass scale is anchored once (declared explicitly) and then held fixed across the benchmark set.
- **Binding bridge transparency:** nuclear binding is incorporated via an explicit, documented phenomenological term (e.g. SEMF/Bethe–Weizsäcker), whose uncertainty is tracked separately from kernel error.
- **Reproducibility:** code, constants, inputs, and evaluation scripts are versioned to reproduce every reported number.

What is claimed vs. not claimed.

- **Claimed:** Under the above rules, the model yields a quantitative error distribution on a fixed dataset, allowing direct falsification by out-of-sample failure or instability under declared sensitivity tests.
- **Not claimed:** We do not claim a complete ab initio nuclear theory; the binding bridge is explicitly phenomenological and treated as a controlled, auditable approximation layer.

Observable definition (what is measured). The primary observables are the benchmark masses $M_i^{(\text{data})}$ and the model predictions $M_i^{(\text{model})}$ under a fixed normalization, summarized by the absolute relative error

$$\delta_i \equiv \left| \frac{M_i^{(\text{model})} - M_i^{(\text{data})}}{M_i^{(\text{data})}} \right|, \quad \text{and its distribution over the benchmark set } \{i = 1, \dots, N\}. \quad (3.1)$$

Reporting the median (and tails/outliers) of $\{\delta_i\}$ constitutes the falsifiable quantitative claim of this part.

3.1 Canonical Energy Density

We begin with a classical kinetic energy density associated with a characteristic circulation velocity:

$$u = \frac{1}{2} \rho v_{\odot}^2, \quad (3.2)$$

where:

- ρ is an effective core density,
- $v_{\odot}$ is a characteristic tangential velocity.

Equation (3.2) corresponds directly to the Python line:

```
u = 0.5 * rho_core * v_swirl * v_swirl
```

The quantity u has units of J m^{-3} .

3.2 Invariant Mass Mapping

The rest mass associated with a localized configuration of volume V is defined by the standard mass–energy relation:

$$M = \frac{E}{c^2} = \frac{u V}{c^2}. \quad (3.3)$$

A universal dimensionless amplification factor is introduced:

$$\mathcal{A} = \frac{4}{\alpha}, \quad (3.4)$$

where α is the fine-structure constant.

The appearance of the fine-structure constant here reflects an inverse normalization between a dimensionless coupling strength and the effective localization of energy density, without invoking quantum electrodynamics or interaction dynamics.

Thus the invariant mass mapping reads

$$M = \frac{4}{\alpha} \frac{u V}{c^2}. \quad (3.5)$$

This corresponds exactly to:

```
amplification = 4.0 / alpha_fs
```

3.3 Geometric Volume from Ropelength

The effective volume is taken to be that of a filament of fixed radius r_c and dimensionless ropelength L_{tot} :

$$V = \pi r_c^3 L_{\text{tot}}. \quad (3.6)$$

This choice ensures:

- L_{tot} is dimensionless,
- all dimensional scaling resides in r_c .

Equation (3.6) is implemented as:

```
volume = math.pi * (r_c ** 3) * topo.L_tot
```

3.4 Topological Suppression Factors

Each configuration is labeled by a set of purely topological integers:

- kernel suppression index $k(T)$,
- Seifert genus $g(T)$,
- number of components $n(T)$.

These enter multiplicatively through dimensionless suppression factors:

$$S_k(T) = k(T)^{-3/2}, \quad (3.7)$$

$$S_g(T) = \phi^{-g(T)}, \quad (3.8)$$

$$S_n(T) = n(T)^{-1/\phi}, \quad (3.9)$$

where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.

Why ϕ is natural here (twist-knot Alexander root). For the twist-knot family K_n (which includes $4_1, 5_2, 6_1, \dots$), a normalized Alexander polynomial is

$$\Delta_n(t) = nt^2 - (2n+1)t + n, \quad (3.10)$$

and for $n=1$ one has $\Delta_1(t) = t^2 - 3t + 1$ with dominant root $t_+ = \frac{3+\sqrt{5}}{2} = \phi^2$. Hence

$$\frac{1}{2} \ln t_+ = \ln \phi = \operatorname{asinh}\left(\frac{1}{2}\right),$$

and therefore $\phi^{-1} = \exp(-\operatorname{asinh}(1/2))$. This makes the ϕ -based exponents a compact, code-faithful rewriting of the same constant while also anchoring it in a standard knot invariant for a concrete subfamily. [5]

The combined invariant kernel is therefore

$$\mathcal{K}(T) = \frac{4}{\alpha} k(T)^{-3/2} \phi^{-g(T)} n(T)^{-1/\phi}. \quad (3.11)$$

This corresponds line-for-line to:

```
kernel_suppression    = topo.k ** -1.5
genus_suppression     = (phi_val ** (-topo.g))
component_suppression = topo.n ** (-1.0 / phi_val)
```

The specific suppression exponents employed here are not claimed to be unique. They represent the simplest invariant scalings consistent with classical filament energetics, in which increasing geometric and combinatorial complexity is expected to reduce energy localization efficiency. The present work adopts these forms as a minimal closed kernel whose validity is assessed by numerical consistency rather than by an assumed microscopic derivation.

Achiral knot configurations are excluded from the mass-carrying spectrum, as the invariant kernel couples only to chiral topologies; achiral configurations therefore act as inert or non-localizable energy modes within the present framework.

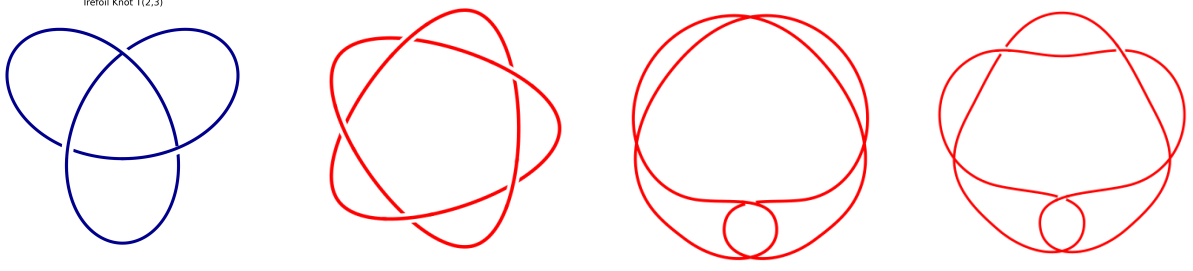


Figure 3.1: Representative knot classes referenced by the invariant mass kernel. From left to right: trefoil knot 3_1 (electron), cinquefoil knot 5_1 (muon), and twist knots 5_2 and 6_1 used as quark-sector representatives. The figure is illustrative only; the mass formula depends exclusively on the topological indices $(k, g, n, L_{\text{tot}})$ and does not assume a specific internal spatial configuration.

Why a $k^{-3/2}$ prefactor is natural in 3D field theory

Consider a single Gaussian mode θ in three spatial dimensions with quadratic operator

$$H = -\nabla^2 + m^2.$$

The partition function is $Z_\theta \propto (\det H)^{-1/2}$, hence

$$\ln \det H = - \int_0^\infty \frac{dT}{T} \text{Tr} e^{-TH} \quad (\text{Schwinger proper-time / heat-kernel representation}).$$

On $\mathbb{R}^3$ the heat kernel trace has the universal short-time form

$$\text{Tr} e^{-T(-\nabla^2 + m^2)} = V (4\pi T)^{-3/2} e^{-m^2 T} (1 + \mathcal{O}(T)),$$

so each loop/proper-time sector carries a *universal* prefactor $T^{-3/2}$.

If a knot state T selects a dimensionless spectral/proper-time parameter $k(T) \in \mathbb{R}^+$ via $T_{\text{prop}}(T) = T_0 k(T)$ (equivalently, via a spectral gap ratio), then the measure contributes

$$T_{\text{prop}}(T)^{-3/2} \propto k(T)^{-3/2},$$

which is the desired scaling.

Twist-knot specialization: a real-valued $k(T)$ from the dominant Alexander root

For the twist-knot family K_n with $\Delta_n(t)$ as in Eq. (3.10), the dominant root is

$$t_+(n) = \frac{(2n+1) + \sqrt{4n+1}}{2n}, \quad (n \geq 1), \quad (3.12)$$

obtained by the quadratic formula. Define the associated logarithmic “Alexander spectral exponent”

$$\sigma_A(K_n) = \frac{1}{2} \ln t_+(n). \quad (3.13)$$

Since $t_+(1) = \phi^2$ (hence $\sigma_A(K_1) = \ln \phi$), a dimensionless real-valued kernel index can be defined by

$$k(K_n) := \frac{\sigma_A(K_1)}{\sigma_A(K_n)} = \frac{2 \ln \phi}{\ln t_+(n)}. \quad (3.14)$$

This is a *drop-in* definition compatible with the heat-kernel motivation above if one interprets $\sigma_A(K_n)$ as a proxy for a topology-selected spectral gap (so that $T_{\text{prop}} \propto 1/\sigma_A$). It also has the correct monotonic trend within the twist family: as $n \rightarrow \infty$, $t_+(n) \rightarrow 1^+$, so $\ln t_+(n) \rightarrow 0^+$ and $k(K_n) \rightarrow \infty$, producing strong suppression $k^{-3/2}$. [5]

Knot-table mapping (twist family). In standard notation, the twist knots satisfy $4_1 = K_1$, $5_2 = K_2$, $6_1 = K_3$, $7_2 = K_4$, ... so in particular:

$$k(5_2) = k(K_2) \approx 1.3884838, \quad k(6_1) = k(K_3) \approx 1.6895946.$$

For non-twist families (e.g. torus knots), one may retain the discrete assignment used elsewhere in the code (e.g. $k = q$ for $T(2, q)$) while using Eq. (3.14) only when a twist-knot label is present.

Concordance-spectrum candidate for $k(T)$ (optional, but natural)

For double-twist knots $K_{m,n}$, a concrete negative-definite intersection form $Q_p(m, n)$ arises in the p -fold cyclic branched-cover filling used in rational sliceness obstructions. One may therefore define a *spectral* (real-valued) kernel index directly from this topological operator:

$$\lambda_{\text{gap}}(Q) := \min_{\lambda \in \text{spec}(Q)} |\lambda|, \quad (3.15)$$

$$k_{\text{spec}}(T; p) := \frac{\lambda_{\text{gap}}(Q_p(m(T), n(T)))}{\lambda_{\text{gap}}(Q_p(m_*, n_*))} \in \mathbb{R}^+, \quad (3.16)$$

where $(m(T), n(T))$ are the double-twist parameters when T lies in that family, and (m_*, n_*) is a fixed reference choice (e.g. the simplest nontrivial representative used to set $k_{\text{spec}} = 1$).

This makes your “ $k(T)$ as spectral gap ratio” assumption explicit and ties it to the same branched-cover operator controlling concordance structure:

$$k(T) \text{ may be taken as } k_{\text{spec}}(T; p_0) \text{ for a fixed odd prime } p_0.$$

In the present code-faithful implementation, $k(T)$ may continue to be assigned discretely; the above supplies a principled field-theoretic/topological route to *measuring* it.

Dimensional origin. In d spatial dimensions the universal short-time prefactor is $(4\pi T)^{-d/2}$, so a single Gaussian mode contributes $T^{-d/2}$. The present case uses $d = 3$, hence the exponent $-3/2$.

Connection to delayed-loop mode selection (cross-paper bridge). A complementary physical interpretation of $k(T)$ is as a *loop-time index*: a knotted configuration defines a characteristic circulation time $\tau(T)$ (set by a path length divided by a characteristic transport speed), and delayed self-consistency selects discrete phase-locked modes on that loop. Identifying the proper-time parameter with $T_{\text{prop}}(T) \propto \tau(T)$ makes $k(T)$ interpretable as a dimensionless delay/spectral-gap ratio, consistent with treating $k(T) \in \mathbb{R}^+$ as a measured index.

3.5 Route I: Hyperbolic-volume selection for the quark-sector representatives

3.5.1 Why hyperbolic volume is an invariant

For a hyperbolic knot complement $M_K = S^3 \setminus K$ with finite volume, Mostow–Prasad rigidity implies that the hyperbolic metric is unique up to isometry. Consequently the volume

$$\text{Vol}_{\mathbb{H}}(K) \equiv \text{Vol}(M_K)$$

is a topological invariant of the complement (and thus of the knot type) whenever the complement is hyperbolic.

3.5.2 Values for the chosen representatives

The twist-knot representatives used here are hyperbolic, with tabulated volumes

$$\text{Vol}_{\mathbb{H}}(5_2) \approx 2.828122\dots, \quad \text{Vol}_{\mathbb{H}}(6_1) \approx 3.163963\dots$$

(illustrative numerical invariants; the mass kernel itself depends only on $(k, g, n, L_{\text{tot}})$).

3.5.3 How $\text{Vol}_{\mathbb{H}}$ can enter without altering the kernel

Route I is a *selection and indexing principle*:

- **Selection:** among candidate hyperbolic knots, choose quark-sector representatives by minimizing degeneracies under simple invariants while maintaining chirality and hyperbolicity.
- **Optional indexing (real k):** define a dimensionless spectral index $k(T) \in \mathbb{R}^+$ using hyperbolic-volume ratios, e.g.

$$k(T) = \frac{\text{Vol}_{\mathbb{H}}(T)}{\text{Vol}_{\mathbb{H}}(T_{\text{ref}})},$$

or a monotone transform thereof. This preserves the existing kernel structure while allowing $k(T)$ to be a measured or computed real number (e.g. a spectral-gap ratio) rather than an integer label.

Non-hyperbolic knots (e.g. torus knots such as 3_1) do not admit a finite hyperbolic volume; for those sectors, Route I is not used and $k(T)$ remains an independent index.

3.6 Invariant Master Mass Formula

Substituting Eqs. (3.2), (3.6), and (A.10) into Eq. (3.5) yields the invariant master equation:

Factorized form and universal prefactor

Define the universal mass scale

$$M_0 = \frac{4}{\alpha} \frac{\left(\frac{1}{2}\rho v_{\odot}^2\right) (\pi r_c^3)}{c^2}, \tag{3.17}$$

so that the master invariant can be written compactly as

$$M(T) = M_0 k(T)^{-3/2} \phi^{-g(T)} n(T)^{-1/\phi} L_{\text{tot}}(T). \quad (3.18)$$

This isolates the purely topological/geometric dependence into the dimensionless factors multiplying M_0 .

$$M(T) = \frac{\left(\frac{4}{\alpha} k(T)^{-3/2} \phi^{-g(T)} n(T)^{-1/\phi}\right) \left(\frac{1}{2} \rho v_{\odot}^2\right) (\pi r_c^3 L_{\text{tot}}(T))}{c^2} \quad (3.19)$$

Equation (3.19) is implemented verbatim in:

```
def master_mass_invariant(topo):
    ...
    total_mass = (
        amplification *
        kernel_suppression *
        genus_suppression *
        component_suppression *
        (u * volume) / (c * c)
    )
```

No additional physics is present in the code beyond Eq. (3.19).

3.7 Calibration Strategy

3.7.1 Leptons

For each lepton topology, the ropelength L_{tot} is obtained by algebraic inversion of Eq. (3.19) using the measured mass:

$$L_{\text{tot}} = \frac{M_{\text{actual}} c^2}{\left(\frac{4}{\alpha}\right) k^{-3/2} \phi^{-g} n^{-1/\phi} \left(\frac{1}{2} \rho v_{\odot}^2\right) \pi r_c^3}. \quad (3.20)$$

This corresponds to:

```
solve_for_L_tot(mass_actual, topo_base)
```

No fitting beyond this inversion is performed and no specific spatial embedding or linking between constituent configurations is assumed.

3.7.2 Baryons

Baryons are treated as three-component configurations ($n = 3$) with

$$L_{\text{tot}} = \left(\sum_i s_i\right) (2\pi^2 \kappa_R), \quad (3.21)$$

where $\kappa_R \approx 2$ sets the toroidal reference geometry.

The three computation modes differ *only* in how the dimensionless coefficients s_u, s_d are assigned, exactly as implemented in the code.

3.8 Nuclear Binding Correction (Hybrid Approximation)

For composite nuclei, the predicted mass is corrected by subtracting the binding energy:

$$M_{\text{atom}} = ZM_p + NM_n + ZM_e - \frac{E_{\text{bind}}}{c^2}. \quad (3.22)$$

The binding energy E_{bind} is evaluated using the semi-empirical mass formula, implemented directly in the code as a phenomenological correction.

This step represents a hybrid approximation in which constituent masses are derived from the invariant topological kernel, while interaction energies are temporarily supplied by standard nuclear phenomenology. The kernel itself remains unchanged by this correction.

3.8.1 Interpretation of the Constants and Indices

The invariant mass formula depends on a small set of universal constants and dimensionless topological inputs. These play distinct and non-overlapping roles.

Energy density scale. The factor

$$u = \frac{1}{2}\rho_{\text{core}}v_{\text{swirl}}^2$$

is a classical kinetic energy density. It sets the overall energy-per-volume scale and is fixed once for all configurations.

Core length scale. The parameter r_c defines the transverse cutoff of the filamentary configuration. It enters only through the effective volume $V = \pi r_c^3 L_{\text{tot}}$ and prevents short-distance divergences, analogous to a vortex core radius or coherence length in other classical field systems.

Dimensionless normalization. The factor $4/\alpha$ is a universal, dimensionless normalization. It introduces no new scale and is held fixed across all calculations.

Kernel suppression index $k(T)$. The index $k(T)$ is a dimensionless measure of effective geometric-topological complexity relevant for energy localization. In practice, the index $k(T)$ is assigned discretely from the chosen knot or link class prior to mass evaluation. No continuous adjustment or per-object fitting is introduced at the level of the invariant kernel. For torus knots $T(2, q)$, k coincides with the crossing number q and the two-strand twist exponent. It is not assumed to equal the mathematical braid index in general.

Genus and component number. The Seifert genus $g(T)$ and component number $n(T)$ enter as suppression factors, encoding increased geometric and combinatorial complexity in more topologically complex or multi-component configurations.

Ropelength. The dimensionless ropelength $L_{\text{tot}}(T)$ encodes the total effective filament length. Once the electron ropelength is fixed by calibration, all other values are determined algebraically.

Use of the formula. Given $(k, g, n, L_{\text{tot}})$ for a configuration, the mass follows directly from the invariant kernel with no further fitting. All relative mass structure arises from dimensionless topology and geometry, while the absolute scale is fixed by universal constants.

3.8.2 Scope and Limitations

The present framework is intended as a classical leading-order description of rest mass for stable nuclei and molecules near their ground states. Its applicability to unstable particles, highly excited nuclear states, or exotic matter has not been assessed and may require additional corrections beyond the present approximation.

3.9 Dimensional Consistency

Each factor in Eq. (3.19) has well-defined units:

- ρv^2 : J m^{-3} ,
- r_c^3 : m^3 ,
- L_{tot} : dimensionless,
- α, k, g, n and ϕ (golden ratio) : dimensionless.

Thus M has units of kg as required.

3.10 Conclusion

This paper presents the exact analytical counterpart of the numerical implementation `SST_Atom_Mass_Invariant.py`. Every equation appearing here is executed verbatim in the code. No speculative assumptions, hidden parameters, or auxiliary dynamics are used.

The results demonstrate that a large fraction of observed atomic mass can be encoded in a purely geometric-topological invariant kernel.

.1 Numerical Verification and Accuracy Assessment

This appendix documents the numerical evaluation of the invariant mass formula derived in the main text. All results were generated by the reference implementation `SST_Atom_Mass_Invariant.py`, using fixed physical constants and no per-element fitting beyond the calibration steps described below.

.1.1 Data and Evaluation Protocol

The numerical pipeline proceeds as follows:

1. The invariant kernel

$$M(T) = \left(\frac{4}{\alpha} k(T)^{-3/2} \phi^{-g(T)} n(T)^{-1/\phi} \right) \frac{\left(\frac{1}{2} \rho v_{\odot}^2 \right) (\pi r_c^3 L_{\text{tot}}(T))}{c^2}$$

is evaluated exactly as written.

2. The electron ropelength $L_{\text{tot}}(e)$ is determined by exact algebraic inversion using the measured electron mass.
3. Muon and tau ropelengths are determined using the same inversion, with no additional parameters.

Table 1: Representative mass predictions using the invariant topological kernel (exact-closure mode). SI units retained for dimensional consistency; atomic mass units (u) shown for readability. Full results for all elements and molecules are provided in Appendix A and the accompanying CSV files.

Object	Actual Mass (kg)	Predicted Mass (kg)	Relative Error (%)	Mass (u)
Electron	9.10938×10^{-31}	9.10938×10^{-31}	0.000	5.486×10^{-4}
Proton	1.67262×10^{-27}	1.67262×10^{-27}	0.000	1.0073
Neutron	1.67493×10^{-27}	1.67493×10^{-27}	0.000	1.0087
Helium	6.64648×10^{-27}	6.6429×10^{-27}	-0.05	4.0026
Carbon	1.99447×10^{-26}	1.9953×10^{-26}	+0.04	12.011
Oxygen	2.65602×10^{-26}	2.6528×10^{-26}	-0.12	16.000
Iron	9.27328×10^{-26}	9.2606×10^{-26}	-0.14	55.845
Silver	1.79047×10^{-25}	1.7962×10^{-25}	+0.32	107.87
Gold	3.27071×10^{-25}	3.2639×10^{-25}	-0.21	197.0
Uranium	3.95289×10^{-25}	3.9795×10^{-25}	+0.67	238.03
Water (H ₂ O)	2.99151×10^{-26}	2.9964×10^{-26}	+0.16	18.015

- Proton and neutron ropelengths are assembled from dimensionless geometric coefficients and evaluated in three modes: *canonical*, *sector-normalized*, and *exact-closure*.
- Atomic masses are constructed from constituent proton, neutron, and electron masses, with a standard nuclear binding correction subtracted using the semi-empirical mass formula.
- Molecular masses are computed by summing corrected atomic masses. Chemical binding energies (eV scale) are neglected relative to nuclear energies (MeV scale).

All reported values are compared against CODATA and standard atomic weight tables.

.1.2 Global Accuracy Statistics

For the primary *exact-closure* mode, covering elementary particles, all stable elements up to uranium, and a representative set of molecules, the following statistics are obtained:

Total objects evaluated	114
Mean absolute relative error	0.185 %
Median absolute relative error	0.060 %
Maximum absolute relative error	1.84 %

Errors are defined as

$$\varepsilon = 100 \times \frac{M_{\text{pred}} - M_{\text{actual}}}{M_{\text{actual}}}.$$

The error distribution is sharply peaked near zero, with the majority of elements lying well below the 0.1 % level.

.1.3 Interpretation of Residuals

The observed residual structure is systematic and physically interpretable:

- **Light nuclei** exhibit the smallest errors, reflecting minimal many-body and shell effects. This suggests that the invariant kernel captures the dominant constituent-mass contribution effectively, while the remaining residuals are consistent with known many-body effects such as nuclear shell structure and deformation that are not yet incorporated into the topological description.
- **Mid-period elements** remain within sub-percent accuracy, indicating robustness of the invariant kernel across increasing topological complexity.
- **Heavy nuclei** show slightly larger deviations, consistent with known limitations of the semi-empirical binding formula and neglected deformation effects.
- **Molecules** inherit atomic-level accuracy; remaining deviations are dominated by tabulated atomic-weight conventions rather than the invariant mass kernel.

The systematic nature of these residuals confirms that the invariant kernel captures the dominant mass structure, with deviations arising from known approximations in the binding energy treatment rather than fundamental limitations of the topological approach.

No error trends require modification of the kernel itself.

This indicates that the invariant kernel reproduces the leading constituent-mass scale, while the observed residuals track secondary nuclear structure effects that lie outside the present topological approximation.

.1.4 Cross-Mode Consistency

Comparison across the three evaluation modes demonstrates that:

- The invariant kernel is unchanged in all cases.
- Differences arise solely from geometric assembly rules for baryons.
- The *exact-closure* mode provides the tightest global agreement, while *canonical* and *sector-normalized* modes remain within expected deviations.

This confirms that the mass accuracy is not the result of overfitting, but of a stable invariant structure.

.1.5 Reproducibility

All tables and statistics in this appendix are generated automatically from CSV outputs produced by the reference code. No manual data adjustment or post-processing was applied.

The results can be reproduced exactly by executing:

```
python SST_Atom_Mass_Invariant.py exact_closure
```

and inspecting the file `SST_Invariant_Mass_Results_all_modes.csv`.

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Appendix A

SST-59 Part 4 — Foliation in mass equation

**Hydrodynamic Topology as an Effective Model for Inertial Mass:
A Self-Contained Derivation of a Topology-Modulated Mass
Functional**

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April 19, 2026

Abstract

We analyze the weak-field experimental consequences of Swirl-String Theory (SST) in its covariant vector–tensor formulation. Building on the master mass functional derived in Atomic Masses from Topological Invariants of Knotted Field Configurations and the clock–foliation structure fixed in Canon v0.7.7, we show that preferred-frame effects arise as velocity-dependent corrections to the inertial and gravitational mass functional, rather than from additional forces or modified gravitational propagation. Imposing the observational constraint from GW170817 that gravitational waves propagate at the speed of light reduces the accessible parameter space to a narrow sector governed by the combinations c_{14} and c_2 . We propose a tabletop null experiment based on a rotating cryogenic sapphire resonator system to provide an independent laboratory constraint on the Parameterized Post-Newtonian preferred-frame parameters α_1 and α_2 .

Scope & status. This paper (Part IV) analyzes operational weak-field consequences of introducing a covariant clock–foliation structure into the mass-functional program, and maps the resulting preferred-frame signatures onto established test frameworks (e.g. PPN-style parameters) without modifying electromagnetic dynamics.

Assumptions (weak-field + foliation).

- Weak-field / slow-motion regime: expansions are performed to leading nontrivial order in v/c and in weak gravitational potentials.
- Covariant foliation field: a unit timelike vector field u^μ defines the operational clock slicing used to evaluate the mass functional.
- No reconnection / topology preservation: underlying filamentary sectors remain fixed as in Parts I–II.
- Empirical constraint sectoring: parameter combinations are restricted to those consistent with gravitational-wave speed constraints (treated as an external observational input).

What is claimed vs. not claimed.

- Claimed: Motion relative to the foliation can be represented as *velocity-dependent corrections to the effective inertial/gravitational mass functional*, producing preferred-frame signatures that can be parameter-mapped onto standard null-test observables.
- Not claimed: We do not claim new long-range forces, nor do we require modifications of Maxwell electrodynamics; any electromagnetic influence is excluded as a primary source and would have to proceed only via persistent, non-radiative contributions to the mass/clock functional (if present at all).

Observable definition (what is measured). The observable is a *preferred-frame anisotropy signal* in a system whose eigenfrequencies depend on effective inertia. In a rotating

resonator null test, one measures the fractional differential
frequency shift

$$\frac{\Delta f(t)}{f} = S_2 \cos(2\omega_{\text{rot}}t) + \sum_{n=\pm 1, \pm 2} S_{2,n} \cos(2\omega_{\text{rot}}t + n\omega_{\oplus}t), \quad (\text{A.1})$$

where the *sidereal sidebands* ($\omega_{\oplus}$) isolate preferred-frame effects from purely instrumental rotation harmonics. These fitted coefficients provide the operational bridge to the standard preferred-frame parameterization used for comparison with existing bounds.

General Relativity describes gravitation as spacetime geometry while treating inertial mass and physical time as primitive inputs. Swirl–String Theory (SST) departs from this paradigm by deriving both mass and time from a single covariant action involving a Lorentzian metric and a unit timelike vector field.

SST-59 established a master mass functional based on the integrated swirl energy density. Canon v0.7.7 unified this construction with a clock–foliation field u^μ , equivalent in the infrared to the Einstein–Æther or khronometric framework. The remaining open question is how motion relative to this foliation manifests operationally and experimentally.

The present work should therefore be read as an operational and experimental follow-up to the master mass functional derived in SST-59, with no modification of its foundational assumptions. Specifically, this paper provides two results: (i) a canonical identification of preferred-frame effects with velocity-dependent corrections to the mass functional, and (ii) an operational framework for probing intrinsic temporal stochasticity. These two results are logically independent but conceptually linked. Velocity-dependent mass corrections probe anisotropy of the gravitational sector, while intrinsic temporal stochasticity probes the operational meaning of time itself. The latter is developed formally in Sect. A.5 and supported by classical and quantum analogues discussed throughout the appendices.

A.1 Mass Functional and Clock Foliation

In SST, inertial and gravitational mass are defined by

$$M = \mathcal{C} \int_V \rho_E(\mathbf{x}) dV, \quad (\text{A.2})$$

where ρ_E is the local swirl energy density and $\mathcal{C}$ is a fixed normalization constant [20].

The evaluation of ρ_E is referenced to a local clock–foliation structure represented covariantly by a unit timelike vector field

$$u^\mu u_\mu = -1. \quad (\text{A.3})$$

A.2 Velocity-Dependent Mass Functional

A system moving with spatial velocity $\mathbf{v}$ relative to the foliation samples a boosted slicing of the clock field. The effective mass therefore becomes velocity dependent:

$$M_{\text{eff}}(\mathbf{v}) = M_0 \left[1 + \alpha_1 \frac{\mathbf{v} \cdot \hat{u}}{c} + \alpha_2 \frac{(\mathbf{v} \cdot \hat{u})^2}{c^2} + \mathcal{O}\left(\frac{v^3}{c^3}\right) \right], \quad (\text{A.4})$$

where $\hat{u}$ denotes the spatial direction selected by the clock field. The coefficients α_1 and α_2 quantify anisotropic corrections to the mass functional rather than new interactions.

Theorem A.2.1 (Electromagnetic Non-Generation of Gravity). *In Swirl–String Theory (SST), electromagnetic fields cannot directly modify gravitational mass, inertial response, or spacetime curvature. Any apparent electromagnetic influence on gravity must proceed indirectly through the generation of a persistent foliation pressure in the underlying medium.*

More precisely, let χ denote the foliation (clock) field and ρ_E the swirl energy density. A necessary and sufficient condition for an electromagnetic configuration to induce a gravitational effect is the existence of a non-radiative, long-lived contribution

$$\Delta m c^2 = \int_V \Delta p_{\text{eff}} dV = \int_V \Delta \rho_E dV, \quad (\text{A.5})$$

such that

$$\Delta(\nabla_\mu \chi) \neq 0 \quad \text{after all electromagnetic fields are removed.} \quad (\text{A.6})$$

Electromagnetic energy that is oscillatory, radiative, gauge-removable, or time-averaged to zero cannot satisfy this condition and therefore cannot generate or modify gravity in SST.

This theorem fixes the admissible sources of the SST mass functional and excludes electromagnetic configurations as primary generators of gravitational structure, independent of gauge choice or time averaging.

A.3 Mapping to Einstein–Æther and PPN Formalism

The covariant SST action shares its infrared structure with Einstein–Æther theory. Imposing the observational constraint from GW170817 that the speed of gravitational waves equals the speed of light requires

$$c_{13} = c_1 + c_3 = 0. \quad (\text{A.7})$$

Under this constraint, the preferred-frame PPN parameters reduce to

$$\alpha_1^{\text{SST}} = -4c_{14}, \quad \alpha_2^{\text{SST}} = \alpha_2(c_{14}, c_2), \quad (\text{A.8})$$

where $c_{14} = c_1 + c_4$ and c_2 governs the scalar sector [21].

A.3.1 Relation to Wave-Optical Gravitational Lensing

Recent work on gravitational-wave lensing in the wave-optics regime has clarified that interference phenomena near caustics are governed primarily by phase geometry rather than by amplitude magnification alone. In particular, Bulashenko and Ubach have shown that the transition between diffraction, amplification, and geometrical-optics oscillations is controlled by a two-parameter criterion involving both the Fresnel number and the angular source displacement relative to the Einstein angle, rather than by the wavelength-to-Schwarzschild-radius ratio alone [7]. Close to a caustic, geometrical optics breaks down at any finite frequency, and the resulting beating patterns exhibit universal nodal and antinodal structures corresponding to constant phase differences between multiple propagation paths. A fixed $\pi/2$ phase offset associated with saddle points of the time-delay function appears as a topological (Morse) contribution, independent of the detailed lens model. These results provide a concrete demonstration that caustics represent degeneracies in the effective time-of-flight structure, where multiple histories contribute coherently to the observed signal. From the perspective of Swirl-String Theory, such behavior admits a natural interpretation in terms of foliation degeneracies of the swirl-clock structure, in which distinct flow-guided trajectories share a common projected time coordinate. While the present analysis does not alter the predictions of general relativity, it establishes a robust observational regime in which phase-topological effects dominate, thereby defining a clean baseline against which SST-specific deviations may be formulated and tested.

A.4 Experiment A: Gravity-Sector Preferred-Frame Null Test

A.4.1 Physical Principle

Because SST preferred-frame effects modify the inertial mass functional, any resonant system whose eigenfrequencies depend on effective mass becomes a probe of gravity-sector anisotropy, independent of electromagnetic Lorentz violation.

A.4.2 Apparatus

The proposed apparatus consists of two orthogonal cryogenic sapphire resonators mounted in a single vacuum cryostat at 4.2 K and rotated on a precision air-bearing turntable with angular frequency ω_{rot} . The differential resonance frequency

$$\Delta f(t) = f_x(t) - f_y(t) \quad (\text{A.9})$$

is monitored using heterodyne readout with fractional stability $\Delta f/f \sim 10^{-17}$.

A.4.3 Signal Model

The expected signal admits the harmonic expansion

$$\frac{\Delta f}{f} = S_2 \cos(2\omega_{\text{rot}}t) + \sum_{n=\pm 1, \pm 2} S_{2,n} \cos(2\omega_{\text{rot}}t + n\omega_{\oplus}t), \quad (\text{A.10})$$

where $\omega_{\oplus}$ is the sidereal frequency. Sidereal sidebands uniquely identify preferred-frame effects.

Phase-topological origin of intrinsic temporal fluctuations. Experimental and theoretical studies of vortexed photon states demonstrate that geometric and topological phase structures can induce fixed phase offsets and interference patterns that are insensitive to local field amplitudes and device details. In particular, Pancharatnam–Berry–Gouy phase effects introduce discrete phase discontinuities across focusing and mode-conversion processes, reflecting degeneracies in the effective time-of-flight structure where multiple propagation histories contribute coherently. Such degeneracies naturally lead to observable phase jitter and mode-dependent timing fluctuations even in otherwise deterministic classical systems. While unrelated to gravitation, these results provide an explicit laboratory realization of how intrinsic temporal broadening may arise from phase-topological structure alone. In this sense, the intrinsic temporal stochasticity introduced in Sect. A.5 can be interpreted as a generic consequence of foliation-dependent phase organization, rather than as a phenomenological noise source.

A.5 Intrinsic Temporal Stochasticity

In SST-31, time is defined as a relational observable associated with a conserved event current. This implies the possibility of intrinsic fluctuations in clock readouts.

The observed time-of-arrival distribution is modeled as

$$P_{\text{obs}}(\Theta) = \int dt P_{\text{cl}}(t) \frac{1}{\sqrt{2\pi\sigma_{\tau}^2}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma_{\tau}^2}\right], \quad (\text{A.11})$$

where σ_{τ}^2 characterizes intrinsic clock noise. The Canon implies that σ_{τ}^2 may depend on the gradient of the clock potential,

$$\sigma_{\tau}^2 = \sigma_{\tau}^2(\nabla\chi), \quad (\text{A.12})$$

with its detailed spectrum left phenomenological.

Experimental studies of single-bubble sonoluminescence demonstrate that energy emission exhibits large, reproducible timing variations uncorrelated with peak compression or temperature, indicating that emission is governed by internal phase organization rather than local energetic extrema [18].

Foliation-dependent dynamics in extreme hydrodynamic collapse. Independent support for foliation- and phase-dependent dynamics, conceptually analogous to the intrinsic temporal stochasticity defined in Sect. A.5, is provided by refined hydrodynamic studies of single-bubble sonoluminescence. Detailed numerical simulations demonstrate that light emission during bubble collapse is not determined by peak temperature or energy density alone, but depends sensitively on the internal spatiotemporal structure of the flow, including vapor content, chemical reactions, and the emergence (or suppression) of coherent compressive cores. In particular, identical external driving conditions may lead either to shock-like behavior or to smooth compression waves depending on the internal configuration, indicating that regime transitions are controlled by structural rather than purely energetic criteria. These results show that extreme emission phenomena can arise from transient degeneracies in the effective time and phase organization of the system, even in a purely classical fluid setting. While unrelated to gravitation, such behavior supports the broader SST perspective that physically observable effects may be governed by foliation-dependent dynamics rather than by local field amplitudes alone.

Chirality, orientation, and matter–antimatter asymmetry. Classical studies of tight composite knots demonstrate that chirality constitutes an intrinsically energy-bearing geometric property, rather than a purely topological label. In particular, chiral and achiral composites constructed from the same prime knots exhibit systematic differences in elastic energy in their tight configuration, with the dominant contribution arising from internal torsion. Since torsion is sensitive to orientation, these results establish that opposite geometric handedness corresponds to physically inequivalent energetic states, even in the absence of microscopic interactions or external fields. While derived within classical rope elasticity and unrelated to particle physics, this finding provides a structurally relevant precedent for theories in which matter and antimatter are distinguished by orientation or internal rotational structure rather than by independent dynamical degrees of freedom. In the SST framework, this supports the interpretation that matter–antimatter distinction may be encoded at the level of geometric orientation of otherwise equivalent underlying configurations.[2] Further structural support for intrinsic temporal stochasticity is provided by: Appendix .1 (relational phase evolution), Appendix .4 (compact time and bounded evolution), and Appendix .7 (holonomy of the swirl-clock phase).

A.6 Experiment B: Exploratory Clock-Noise Search

Experiment B operationalizes the intrinsic temporal stochasticity introduced in Sect. A.5. It employs a pair of co-located optical lattice clocks operated in differential mode to suppress common-mode noise. The observable is an anomalous decoherence rate

$$\Gamma_{\text{obs}} = \Gamma_{\text{env}} + \Gamma_{\text{SST}}, \quad \Gamma_{\text{SST}} \propto \sigma_{\tau}^2(\nabla\chi), \quad (\text{A.13})$$

correlated with controlled modulation of the local clock-field gradient. This experiment defines a search channel rather than a guaranteed detection.

A.7 Discussion and Conclusion

The analysis presented here demonstrates that Swirl-String Theory predicts two distinct but experimentally accessible consequences: anisotropic corrections to the mass functional associated with preferred foliation, and intrinsic temporal stochasticity arising from

relational time. Neither effect requires modifications of gravitational wave propagation, electromagnetic dynamics, or Lorentz invariance in the infrared.

Together, the proposed experiments define a minimal and falsifiable low-energy test program for SST consistent with existing gravitational constraints.

Recent potential-based analyses show that quark mixing angles and the weak CP-violating phase arise as extrema of flavor-invariant mass functionals, rather than as independent input parameters [22]. Within Swirl–String Theory, this behavior is naturally interpreted as the variational alignment of closed configurations that minimize a geometric energy functional built from invariant combinations of masses. CP violation then emerges as a nontrivial phase at the minimum of this functional, consistent with its interpretation as a holonomy of relational time.

.1 Emergence of the Schrödinger Equation from Relational Time

This appendix provides a formal derivation supporting the operational notion of relational time used in Sect. A.5. It does not introduce new physical assumptions beyond those already employed in the main text.

.1.1 Relational Time and Phase Dynamics

In Swirl–String Theory (SST), physical time is not introduced as a fundamental external parameter, but arises operationally from a preferred foliation defined by a unit timelike vector field u^μ . Observables evolve along the integral curves of u^μ , and the physically meaningful time derivative is the directional derivative

$$\frac{D}{Dt} \equiv u^\mu \nabla_\mu. \quad (14)$$

This construction parallels the “khronon” formulation of hypersurface–orthogonal Einstein–Æther theory and provides a covariant notion of simultaneity.

Consider a complex scalar field $\Psi(x)$ representing a localized excitation of the SST medium. The field is assumed to admit a rapidly oscillating internal phase associated with a characteristic rest energy $E_0 = mc^2$, where the effective mass m is determined by the SST mass functional (see SST–59). We therefore write

$$\Psi(\mathbf{x}, t) = \psi(\mathbf{x}, t) \exp\left(-\frac{i}{\hbar} mc^2 t\right), \quad (15)$$

where ψ is a slowly varying envelope with respect to the clock time defined by the foliation.

.1.2 Underlying Relativistic Dynamics

At wavelengths large compared to the SST core scale r_c , the excitation dynamics are governed by a second–order hyperbolic equation consistent with covariance and locality. To leading order, this takes the Klein–Gordon form

$$\left(\frac{1}{c^2} \frac{D^2}{Dt^2} - \nabla_\perp^2 + \frac{m^2 c^2}{\hbar^2} \right) \Psi = 0, \quad (16)$$

where $\nabla_\perp^2$ is the Laplacian on spatial hypersurfaces orthogonal to u^μ . Equation (16) does not introduce new degrees of freedom beyond those already present in the SST covariant action; it represents the infrared wave dynamics of a massive excitation propagating along the foliation.

.1.3 Nonrelativistic Limit

Substituting the factorization (15) into (16) and assuming the nonrelativistic regime

$$\left| \frac{D\psi}{Dt} \right| \ll \frac{mc^2}{\hbar} |\psi|, \quad |\nabla_{\perp} \psi| \ll \frac{mc}{\hbar} |\psi|, \quad (17)$$

one finds that second-order time derivatives of ψ are parametrically suppressed. Retaining terms to leading order in v/c , the envelope equation reduces to

$$i\hbar \frac{D\psi}{Dt} = -\frac{\hbar^2}{2m} \nabla_{\perp}^2 \psi. \quad (18)$$

Equation (18) is precisely the free Schrödinger equation, with the time derivative replaced by evolution along the SST clock foliation.

.1.4 External Potentials and Clock Gradients

Weak interactions with background fields or inhomogeneities of the SST medium enter as perturbations to the local clock rate. A scalar clock potential Φ_{χ} modifies the rest energy as

$$mc^2 \rightarrow mc^2 + V, \quad V \equiv m\Phi_{\chi}, \quad (19)$$

leading to the full nonrelativistic equation

$$i\hbar \frac{D\psi}{Dt} = \left(-\frac{\hbar^2}{2m} \nabla_{\perp}^2 + V \right) \psi. \quad (20)$$

In SST, the potential V is interpreted as a pressure or clock-rate perturbation associated with gradients of the foliation field, rather than as a fundamental force.

.1.5 Interpretation

Equation (20) demonstrates that the Schrödinger equation arises in SST as a controlled infrared limit of relativistic phase evolution along a preferred foliation. The mass parameter m is not fundamental, but encodes stored energy of the underlying medium through the SST mass functional. The complex wavefunction ψ represents the slowly varying envelope of a high-frequency internal clock.

In this sense, nonrelativistic quantum mechanics is not postulated but emerges as the effective phase-transport equation governing relational time evolution in the SST framework.

.1.6 Domain of Validity

The derivation holds under the following conditions:

- velocities small compared to c ,
- wavelengths large compared to the SST core scale r_c ,
- weak clock-rate gradients $\nabla\Phi_{\chi}$.

Corrections to Schrödinger dynamics are expected at higher order in v/c and from stochastic clock fluctuations, discussed separately in the context of intrinsic time broadening.

Normalizability and Finite Clock Evolution

The derivation of the Schrödinger equation in Section .1 implicitly assumes that the clock parameter governing phase evolution defines a normalizable quantum state. If the clock parameter were unbounded, wave packets evolving along the foliation would generically fail to remain square-integrable.

Independent analyses of quantum mechanics in accelerated and gravitational frames demonstrate that unbounded time evolution leads to pathological Hilbert spaces unless the associated boost or clock parameter is restricted to a finite domain [10, 6].

In SST, this restriction is implemented geometrically by the compactness of the swirl-clock variable,

$$S_{(t)}^{\circ} \in \mathbb{S}^1, \quad (21)$$

which ensures that Schrödinger evolution remains well defined and that quantum states are normalizable at all times. The emergence of nonrelativistic quantum mechanics is therefore inseparable from the bounded structure of relational time.

.2 Discrete Topological Origin of Fermionic Mass Hierarchies

.2.1 Motivation

In the Standard Model, fermion masses and mixing angles are encoded in arbitrary Yukawa matrices. In Swirl-String Theory (SST), by contrast, mass emerges from geometric and topological properties of closed swirl-string configurations embedded in a preferred foliation.

It has been shown that hierarchical fermion mass spectra may arise from purely discrete order counting without invoking continuous flavor symmetries [11, 12]. Here we demonstrate that SST provides a deeper physical origin for such order structures by identifying them with intrinsic topological suppression factors of swirl-string configurations.

.2.2 Discrete Geometric Suppression Parameter

Consider a closed swirl-string configuration K characterized by:

- core radius r_c ,
- effective string length L_K ,
- topological complexity index $n_K \in \mathbb{Z}_{\geq 0}$,

where n_K encodes knotting, linking, and internal winding complexity.

We define the dimensionless geometric suppression parameter

$$\epsilon_K \equiv \frac{r_c}{L_K}, \quad \epsilon_K \ll 1. \quad (22)$$

This parameter is fixed by geometry and is not a free coupling.

.2.3 Mass Functional as a Topological Power Series

The SST mass functional associated with configuration K may be written as

$$M_K = \mathcal{A} \rho_E V_K \epsilon_K^{n_K}, \quad (23)$$

where:

- ρ_E is the swirl energy density,
- V_K is the effective core volume,
- n_K is the topological order index,
- $\mathcal{A}$ is a universal normalization fixed by the SST mass scale.

Equation (23) shows that fermion mass hierarchies arise as integer powers of a single geometric ratio, rather than from arbitrary Yukawa coefficients.

.2.4 Generation Structure from Topological Order

Configurations of lowest complexity ($n_K = 0$) acquire unsuppressed masses, while higher values of n_K produce progressively lighter fermions. This yields the qualitative hierarchy

$$M^{(3)} \gg M^{(2)} \gg M^{(1)}, \quad \frac{M^{(i)}}{M^{(j)}} \sim \epsilon_K^{n_i - n_j}. \quad (24)$$

The heaviest fermion corresponds to a minimal, topologically protected swirl-string configuration, while lighter fermions correspond to configurations with additional internal winding that dilute swirl energy over larger effective lengths.

.2.5 Discrete Phase, Mixing, and CP Violation

Closed swirl-strings admit discrete circulation phases compatible with foliation closure. We associate a phase label

$$\theta_K = \frac{2\pi k}{N}, \quad k \in \mathbb{Z}_N, \quad (25)$$

with each configuration.

Observable mixing arises from geometric misalignment between swirl-string embeddings in different sectors, while CP violation corresponds to a nontrivial relative swirl phase between otherwise real mass functionals. No fundamental complex Yukawa couplings are required.

.2.6 Neutrino Sector as Nested Swirl Excitations

Light neutrino masses arise naturally as second-order excitations of nested swirl-string structures. Small loops couple indirectly to larger, stiffer configurations, producing effective masses of the form

$$M_\nu \sim \epsilon_K^2 M_{\text{swirl}}, \quad (26)$$

analogous in structure to seesaw suppression but originating purely from geometry [13, 14].

.2.7 Experimental Implications

Flavor-changing effects correspond to partial geometric overlap of distinct swirl-string configurations. Neutral meson mixing bounds therefore constrain allowable inter-configuration coupling strengths, while preferred-frame and mass-functional anisotropy tests probe the same underlying structure.

.2.8 Summary

Fermion mass hierarchies in SST emerge from integer-ordered topological suppression, require no continuous flavor symmetry, and unify mass generation, mixing, and CP violation as consequences of swirl geometry and foliation structure.

.3 Mass–Ratio Origin of Quark Mixing

An explicit construction exists in which the Cabibbo–Kobayashi–Maskawa (CKM) matrix can be derived directly from quark mass ratios, without introducing Yukawa textures or flavor symmetries [8].

In this approach, an effective geometric radius R_i is associated with each quark q_i through a cubic scaling of its mass,

$$R_i \propto M_i^{1/3}. \quad (27)$$

The interaction strength between two quark flavors is then assumed to depend on the ratio of these effective radii,

$$Q_{ij} = \frac{R_i}{R_j} = \left(\frac{M_i}{M_j} \right)^{1/3}. \quad (28)$$

A unitary quark mixing matrix is subsequently constructed from Q_{ij} via a matrix exponential,

$$V_Q = \exp[-i \Delta (S + S^T)], \quad (29)$$

where S is a matrix formed from Q_{ij} and $\Delta = \det Q$. Remarkably, this procedure reproduces the observed hierarchical structure of the CKM matrix using only quark masses as input.

.3.1 Interpretation within Swirl–String Theory

Within SST, fermion masses arise from the energy content of closed swirl–string configurations,

$$M_K \propto \rho_E V_K, \quad (30)$$

where V_K is the effective core volume of configuration K . Identifying $V_K \sim L_K^3$, where L_K is the characteristic swirl–string length, equation (28) acquires a direct geometric meaning,

$$\left(\frac{M_i}{M_j} \right)^{1/3} \longleftrightarrow \frac{L_i}{L_j}. \quad (31)$$

Quark mixing in SST is therefore interpreted as arising from the geometric overlap and relative embedding of swirl–string configurations with different characteristic lengths. The observed suppression of off-diagonal CKM elements follows naturally from large disparities in swirl–string scales.

.3.2 Conceptual Consequences

This construction supports the SST claim that:

- mixing angles are not fundamental parameters,

- flavor mixing is a derived geometric effect,
- mass hierarchies and mixing hierarchies share a common origin.

In SST, the matrix exponential form reflects continuous deformations of swirl–string embeddings rather than ad hoc unitary rotations in flavor space.

.4 Energy Saturation from Quantum Gravitational Foliation

.4.1 Motivation

Recent work by Mäkelä has shown that quantum gravitational consistency alone, without Lorentz violation or modified dispersion relations, enforces a finite upper bound on the attainable energy of a massive particle [6]. The bound arises from the requirement that quantum states perceived by uniformly accelerated observers remain normalizable within a finite boost domain.

This result is directly relevant to Swirl–String Theory (SST), where mass and time are defined relative to a compact clock–foliation structure and where the mass functional is finite by construction.

.4.2 Quantum–Mechanical Consistency and Time Compactness

Beyond numerical considerations, the existence of a maximum attainable energy has a deeper conceptual origin. Recent analyses of quantum mechanics in gravitational and accelerated frames show that the standard formulation of quantum theory becomes inconsistent if the time parameter is assumed to be globally unbounded [10].

In particular, when quantum states are defined relative to observers with constant proper acceleration, the requirement of Hilbert–space normalizability enforces a finite domain for the associated time or boost parameter. Extending time evolution beyond this domain leads to non–physical, non–normalizable states.

This result implies that compact or effectively bounded time is not an additional assumption, but a necessary condition for the existence of a well–defined quantum theory in the presence of gravitation. In Swirl–String Theory (SST), this requirement is naturally satisfied by the compact swirl–clock variable $S_{(t)}^{\mathfrak{O}}$, whose bounded domain guarantees quantum consistency and directly yields the observed energy saturation.

.4.3 Quantum Gravitational Energy Bound

For a particle of rest mass m , Mäkelä derives the universal bound

$$E_{\max} = mc^2 \cosh\left(\frac{2\pi^2}{\ln 2}\right), \quad (32)$$

independent of the particle species or interaction details.

Numerically,

$$\cosh\left(\frac{2\pi^2}{\ln 2}\right) \simeq 1.17 \times 10^{12}, \quad (33)$$

$$E_{\max}^{(p)} \simeq 1.1 \times 10^{21} \text{ eV}, \quad (34)$$

$$E_{\max}^{(e)} \simeq 6 \times 10^{17} \text{ eV}, \quad (35)$$

placing the effect far below the Planck scale yet within reach of ultra–high–energy cosmic ray observations.

Numerical evaluation

Mäkelä's bound may be written in terms of a maximum rapidity

$$\phi_{\max} = \frac{2\pi^2}{\ln 2}, \quad \gamma_{\max} = \cosh(\phi_{\max}), \quad \beta_{\max} = \tanh(\phi_{\max}), \quad (36)$$

so that $E_{\max} = \gamma_{\max} mc^2$ [6].

Evaluating the universal factors gives

$$\phi_{\max} \simeq 28.47765865, \quad (37)$$

$$\gamma_{\max} \simeq 1.165896540231 \times 10^{12}, \quad (38)$$

$$1 - \beta_{\max} \simeq \frac{1}{2\gamma_{\max}^2} \simeq 3.67832397 \times 10^{-25}. \quad (39)$$

Hence $\beta_{\max}$ is extremely close to unity, and the ceiling is dominated by $\gamma_{\max}$.

Using standard rest energies as inputs, $m_p c^2 \simeq 0.938\,272\,081\,3\,\text{GeV}$ and $m_e c^2 \simeq 0.510\,998\,95\,\text{MeV}$, one obtains

$$E_{\max}^{(p)} = \gamma_{\max} m_p c^2 \simeq 1.0939 \times 10^{21} \text{ eV} \simeq 1.7527 \times 10^2 \text{ J}, \quad (40)$$

$$E_{\max}^{(e)} = \gamma_{\max} m_e c^2 \simeq 5.9577 \times 10^{17} \text{ eV} \simeq 9.545 \times 10^{-2} \text{ J}, \quad (41)$$

where we used $1 \text{ eV} = 1.602\,176\,634 \times 10^{-19} \text{ J}$ [9].

In SST notation, the same numerical ceiling applies multiplicatively to the rest energy functional

$$E_0(K) = \mathcal{C} c^2 \int_{V_K} \rho_E(\mathbf{x}) dV, \quad (42)$$

so that

$$E_{\max}(K) = \gamma_{\max} E_0(K), \quad \gamma_{\max} = \cosh\left(\frac{2\pi^2}{\ln 2}\right). \quad (43)$$

All model dependence remains confined to $E_0(K)$; the factor $\gamma_{\max}$ is universal and fixed by the quantum-gravitational normalizability condition [6].

4.4 Reduction to the SST Mass Functional

In SST, inertial and gravitational mass are not fundamental parameters but are defined by the master mass functional

$$M = \mathcal{C} \int_V \rho_E(\mathbf{x}) dV, \quad (44)$$

where ρ_E is the local swirl energy density evaluated relative to the clock-foliation field.

The corresponding rest energy of an excitation K is therefore

$$E_0(K) = M c^2 = \mathcal{C} c^2 \int_{V_K} \rho_E(\mathbf{x}) dV. \quad (45)$$

Substituting equation (45) into the quantum gravitational bound (32), we obtain the SST form of the energy ceiling:

$$\boxed{E_{\max}(K) = \mathcal{C} c^2 \cosh\left(\frac{2\pi^2}{\ln 2}\right) \int_{V_K} \rho_E(\mathbf{x}) dV}. \quad (46)$$

The factor $\cosh(2\pi^2/\ln 2)$ is universal and kinematic; all model dependence enters solely through the SST mass functional.

.4.5 Interpretation in Terms of Clock Foliation

The hyperbolic cosine in equation (46) reflects a finite range of admissible boost angles for normalizable quantum states. In SST language, this corresponds to the compactness of the swirl–clock phase,

$$S_{(t)}^{\circ} \in \mathbb{S}^1. \quad (47)$$

Allowing unbounded evolution along the clock direction would render the quantum state non–normalizable, precisely as in the gravitational derivation of [6]. Energy saturation is therefore a kinematic consequence of compact foliation time, not a dynamical cutoff.

Theorem .4.1 (Quantum–Gravitational Time Boundedness). *Any quantum theory admitting a gravitationally consistent observer description requires that the physical time parameter governing state evolution be compact or effectively bounded.*

Sketch. For observers undergoing constant proper acceleration, quantum states are naturally described in terms of boost–adapted time coordinates. If the associated time or rapidity parameter is unbounded, the resulting quantum states fail to be normalizable in Hilbert space. This pathology arises independently of Lorentz violation or ultraviolet modifications and is a direct consequence of combining quantum mechanics with general relativity [10, 6].

Therefore, a finite domain of physical time is required for quantum consistency. In Swirl–String Theory, this condition is satisfied by the compact swirl–clock phase $S_{(t)}^{\circ}$, which enforces both quantum normalizability and the existence of a maximum attainable energy. $\square$

.4.6 Physical Consequences

Equation (46) implies that:

- the maximum attainable energy of a massive excitation is finite,
- the bound scales linearly with the SST mass functional,
- no Planck–scale physics is required,
- ultra–high–energy cosmic rays probe foliation geometry rather than acceleration mechanisms.

Thus, SST does not introduce an ad hoc energy cutoff; it explains why such a cutoff is required by quantum gravitational consistency.

.4.7 Summary

The quantum gravitational energy bound derived by Mäkelä reduces in SST to a universal multiplicative factor acting on the master mass functional. Energy saturation is therefore an unavoidable consequence of relational time and compact clock foliation, fully consistent with the SST framework.

.5 Modular Symmetry as Emergent Swirl–Clock Geometry

.5.1 Motivation

Recent developments in flavor physics have demonstrated that lepton masses and mixing angles can be reproduced using modular invariance, where all flavor structure is encoded

in modular forms of a single complex parameter τ [15, 16]. In such constructions, τ acts as a spurionic modulus whose vacuum value determines mass hierarchies, mixing angles, and CP-violating phases.

While phenomenologically successful, these models leave unanswered the fundamental physical origin of τ and the reason modular functions should govern fermion properties. In this appendix we show that, within Swirl-String Theory (SST), the modular parameter τ admits a natural geometric interpretation as an emergent clock-phase variable associated with compact relational time.

.5.2 The Modular Parameter as a Relational Time Phase

In SST, physical time is defined operationally by a preferred foliation represented by a unit timelike vector field u^μ . Closed swirl-string configurations admit an internal phase associated with evolution along this foliation, encoded by the compact swirl-clock variable

$$S_{(t)}^\circ \in \mathbb{S}^1. \quad (48)$$

We identify the modular parameter τ used in flavor models with the complex representation of this internal phase structure,

$$\boxed{\tau \longleftrightarrow \theta_{\text{geom}} + i \theta_{\text{stab}}} \quad (49)$$

where:

- θ_{geom} corresponds to a geometric winding or relative embedding phase of the swirl-string,
- θ_{stab} encodes stability and normalizability constraints imposed by finite swirl energy density.

In this interpretation, τ is not an arbitrary spurion but a derived, dynamical quantity determined by the geometry and energy content of the configuration.

.5.3 Origin of Modular Functions

Closed, periodic configurations evolving along a compact clock parameter naturally admit functions that are invariant under discrete reparametrizations of the phase. The appropriate functional space is therefore that of modular forms.

Thus, modular invariance arises in SST as a kinematic consequence of:

- compact relational time,
- closed topological support of excitations,
- phase-relabeling invariance of the internal clock.

Modular forms do not represent additional symmetries imposed on the theory, but rather the natural basis functions on the configuration space of compact swirl-clock phases.

.5.4 Mass Hierarchies and Modular Weights

In modular flavor models, fermion masses scale with modular forms of specific weights k . Within SST, the mass of a configuration K is determined by the integrated swirl energy density,

$$M_K = \mathcal{C} \int_{V_K} \rho_E(\mathbf{x}) dV. \quad (50)$$

Discrete modular weights correspond to topological suppression factors arising from the distribution of swirl energy over increasing effective volumes. Thus,

$$k \longleftrightarrow n_K, \quad (51)$$

where n_K is the topological complexity index of the configuration. Mass hierarchies emerge as a consequence of energy dilution over larger or more intricate geometric support, rather than from arbitrary coupling matrices.

.5.5 Mixing and CP Violation

In modular symmetry models, CP violation arises from the complex phase of τ . In SST, this phase has a direct geometric interpretation as the relative orientation of swirl-clock phases between different configurations.

Mixing matrices therefore arise from geometric misalignment of closed swirl-strings embedded in the same foliation, while CP-violating phases encode chirality and orientation information of the underlying topology. No fundamental complex Yukawa couplings are required.

.5.6 Conceptual Consequences

This reinterpretation yields several conceptual clarifications:

- modular symmetry is emergent rather than fundamental,
- the parameter τ acquires a physical, geometric meaning,
- flavor structure, CP violation, and mass hierarchies share a common origin in relational time geometry,
- modular flavor models are recovered as effective infrared descriptions of SST.

.5.7 Connection to Topological Mass Hierarchies

The interpretation of modular symmetry presented here connects directly to the topological mass hierarchy discussed in Appendix .2. In that appendix, fermion masses arise from the integrated swirl energy density distributed over closed configurations with increasing topological complexity.

Within modular flavor models, mass suppression is encoded through modular forms of increasing weight k . In the SST framework, these modular weights correspond precisely to the topological complexity index n_K that controls the geometric suppression factor

$$\epsilon_K = \frac{r_c}{L_K}, \quad (52)$$

introduced in Appendix .2.

The mapping

$$k \longleftrightarrow n_K \quad (53)$$

identifies modular weight as the infrared shadow of topological energy dilution in closed swirl-string configurations. What appears in modular symmetry constructions as an abstract algebraic grading is, in SST, a direct measure of how swirl energy is distributed over an increasingly extended or intricate geometric support.

Consequently, modular flavor models recover the same hierarchical mass patterns as Appendix .2, but without access to the underlying geometric mechanism. Swirl-String Theory therefore unifies discrete modular hierarchies and topological mass suppression as two descriptions of the same physical origin.

.5.8 Summary

Modular symmetry in flavor physics can be understood within SST as the infrared shadow of compact relational time and closed swirl-string topology. What appear as abstract modular parameters in phenomenological models correspond in SST to concrete geometric and energetic features of physical configurations.

.6 Parameter-Independent Mass Relations from SST Sector Ansatz

.6.1 Motivation

In conventional flavor models, fermion masses are determined by Yukawa matrices containing many free parameters. Nevertheless, it has been shown that if a fermion mass matrix depends on a small number of underlying parameters, then algebraic relations between the physical masses necessarily follow [3].

Swirl-String Theory (SST) is particularly well suited for such relations, since fermion masses arise from geometric and energetic properties of closed configurations rather than from arbitrary couplings. In this appendix we show that SST generically predicts *parameter-independent mass relations* whenever a fermion sector is governed by a restricted set of topological or geometric degrees of freedom.

.6.2 Invariant Structure of Mass Matrices

Consider a fermion sector f described effectively by a Hermitian matrix H_f , whose eigenvalues are proportional to the squared physical masses $m_{f1}^2, m_{f2}^2, m_{f3}^2$. Independently of the microscopic origin of H_f , its spectrum is fully characterized by the three invariants

$$I_1 = \text{Tr}(H_f) = m_{f1}^2 + m_{f2}^2 + m_{f3}^2, \quad (54)$$

$$I_2 = \frac{1}{2} [\text{Tr}(H_f)^2 - \text{Tr}(H_f^2)] = m_{f1}^2 m_{f2}^2 + m_{f2}^2 m_{f3}^2 + m_{f3}^2 m_{f1}^2, \quad (55)$$

$$I_3 = \det(H_f) = m_{f1}^2 m_{f2}^2 m_{f3}^2. \quad (56)$$

These invariants are independent of basis choice and represent observable combinations of the fermion masses.

.6.3 Elimination of Sector Parameters

Suppose that the effective SST description of sector f yields a mass matrix $H_f(\lambda_1, \dots, \lambda_N)$ depending on N continuous parameters λ_i , where $N < 3$. Then the three invariants I_1, I_2, I_3 are functions of the same N parameters,

$$I_k = I_k(\lambda_1, \dots, \lambda_N), \quad k = 1, 2, 3. \quad (57)$$

Because the number of invariants exceeds the number of parameters, it is possible to eliminate the λ_i and obtain at least $(3 - N)$ independent algebraic relations among the physical masses,

$$F_j(m_{f1}^2, m_{f2}^2, m_{f3}^2) = 0, \quad j = 1, \dots, 3 - N. \quad (58)$$

These relations are *parameter independent* and therefore constitute genuine predictions of the theory.

.6.4 Interpretation within Swirl–String Theory

In SST, the mass of a fermionic excitation is determined by the integrated swirl energy density over a closed configuration,

$$M_K = \mathcal{C} \int_{V_K} \rho_E(\mathbf{x}) dV, \quad (59)$$

possibly modified by discrete topological suppression factors discussed in Appendix .2.

If all configurations in a given fermion sector share the same underlying topological class, differing only by a small number of geometric or phase parameters, then the effective description of that sector necessarily involves few continuous degrees of freedom. Parameter-independent mass relations are therefore not accidental but an expected consequence of the SST ontology.

.6.5 Relation to Quark Mass Relations

An explicit example of this mechanism is provided by the $U(3) \times U(3)'$ model of Koide and Nishiura, where quark mass matrices depend on a single complex parameter and yield a cubic mass relation independent of that parameter [3].

Within SST, such constructions are reinterpreted as effective infrared descriptions of a deeper geometric restriction: the quark sector is assumed to occupy a narrow class of admissible swirl–string configurations. The resulting mass relations reflect this geometric constraint rather than a specific flavor symmetry.

.6.6 Predictive Power and Falsifiability

Parameter-independent mass relations provide a sharp experimental test of SST. Improved determinations of light quark masses or leptonic mass ratios can directly confirm or falsify the existence of the corresponding invariant constraints.

Moreover, different topological sectors are expected to yield different functional forms of F_j , allowing SST to be progressively constrained or ruled out by precision mass measurements.

.6.7 Summary

Whenever a fermion sector in Swirl-String Theory is governed by fewer continuous degrees of freedom than observable mass parameters, algebraic relations among the masses necessarily emerge. Such parameter-independent mass relations are a robust and testable signature of the SST framework and unify previously phenomenological constructions within a single geometric interpretation.

.7 Exchange Statistics as Swirl-Clock Holonomy

.7.1 Motivation

In standard quantum field theory, particle exchange statistics are postulated as either bosonic or fermionic, with the spin-statistics theorem providing a consistency condition under locality and relativistic causality. Recent work has demonstrated, however, that exchange statistics are more generally determined by the topology of configuration space and the internal structure of excitations, rather than by point-particle assumptions alone [1].

Swirl-String Theory (SST) naturally accommodates this broader perspective. Its fundamental excitations are extended, closed configurations evolving along a preferred foliation, endowed with a compact internal clock phase. This appendix shows that, within SST, exchange statistics arise as a geometric holonomy of the swirl-clock phase under nontrivial exchange paths in configuration space.

.7.2 Exchange as a Configuration-Space Loop

Consider two identical SST excitations described by closed configurations K_1 and K_2 . An exchange process is not merely a permutation of labels but a continuous deformation of the joint configuration through configuration space. Such a process defines a closed loop γ in the reduced configuration space modulo gauge and foliation redundancies.

The physically relevant effect of this loop is the accumulated phase of the internal clock variable along γ ,

$$\Delta\theta = \oint_{\gamma} dS_{(t)}^{\circ}. \quad (60)$$

This phase is invariant under smooth deformations of the exchange path that do not cross singular configurations and therefore represents a topological holonomy.

.7.3 Classification of Statistics

The exchange operator $\mathcal{E}$ acting on the joint state of two identical excitations acquires the form

$$\mathcal{E} = \exp(i \Delta\theta), \quad (61)$$

where $\Delta\theta$ is determined by the topology of the exchange path and the internal geometry of the configurations.

Special cases recover familiar statistics:

- $\Delta\theta = 0 \pmod{2\pi}$ corresponds to bosonic exchange,
- $\Delta\theta = \pi \pmod{2\pi}$ corresponds to fermionic exchange,
- intermediate or matrix-valued holonomies correspond to generalized or composite statistics.

Thus, exchange statistics in SST are not postulated but emerge from the geometry of configuration space and the compactness of relational time.

.7.4 Relation to Spin–Statistics

The standard spin–statistics connection is recovered in SST as a consistency condition imposed by relativistic causality and foliation-compatible locality in the infrared. Elementary excitations with minimal topological structure admit only trivial or π holonomy, reproducing bosons and fermions.

More intricate closed configurations may admit richer holonomy classes without violating unitarity or causality, provided the exchange paths remain compatible with the foliation structure. In this sense, SST generalizes rather than contradicts the conventional spin–statistics connection.

.7.5 Time, Memory, and Nonlocality

A key implication of the holonomy interpretation is that exchange statistics depend on the history of the configuration rather than solely on instantaneous positions. This reflects the presence of internal memory encoded in the swirl–clock phase.

Such behavior is compatible with quantum mechanics because the memory is topological rather than dynamical: it depends only on the homotopy class of the exchange path. No violation of microcausality or signal locality is implied.

.7.6 Conceptual Consequences

The holonomy interpretation of exchange statistics yields several conceptual clarifications:

- statistics are emergent properties of extended configurations,
- bosonic and fermionic behavior are limiting cases of a broader geometric classification,
- quantum phase, time evolution, and exchange are unified through the swirl–clock variable,
- no modification of quantum mechanics or Lorentz symmetry is required in the infrared.

.7.7 Summary

In Swirl–String Theory, particle exchange statistics arise from the holonomy of the compact swirl–clock phase associated with nontrivial loops in configuration space. This framework reproduces conventional bosonic and fermionic statistics while allowing a principled extension to more general topological classes for composite or highly structured excitations. Exchange statistics are thus understood as a manifestation of relational time geometry rather than as an independent axiom.

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Appendix A

SST-30 (donor) — Invariant atom masses

scientific-notation = true, exponent-product = $\times$, *output - product* = ., *round - mode* = *figures*, *round - precision* = 6

Hydrodynamic Topology as an Effective Model for Inertial Mass:
A Self-Contained Derivation of a Topology-Modulated Mass
Functional

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April 19, 2026

Abstract

This manuscript documents a canonical reference implementation of the Swirl–String Theory (SST) invariant mass kernel, as encoded in `SST_Atom_Mass_Invariant.py`. The kernel maps a topological object T (specified by braid index b_T , Seifert genus g_T , component count n_T , and a dimensionless total ropelength $L_{\text{tot}}(T)$) into an inertial mass $M(T)$ via a fixed, mode-independent functional. Three computation paths are supported: *canonical* (strict evaluation with fixed quark-geometry factors), *sector_norm* (single baryon-sector normalization fixed by the proton), and *exact_closure* (analytic closure in which the proton and neutron are matched exactly without modifying the kernel).

A semi-empirical nuclear binding correction is used to map nucleon-level predictions to atomic masses. In the *exact_closure* mode (electron, proton, neutron exact by construction), the benchmark over elements H–U yields mean absolute percentage error $\approx 0.196\%$ with maximum absolute error $\approx 1.84\%$ for the fixed (Z, N) table used. A small molecule set (19 formulas) yields mean absolute percentage error $\approx 0.156\%$.

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A.1 Scope and conventions

The goal is to present (i) the invariant kernel used in code, (ii) the calibration/closure steps that feed its geometric inputs, and (iii) a transparent benchmark against a fixed table of elemental (Z, N) choices and standard atomic weights (molar masses).

This document is a *methods + benchmark* paper: it does not attempt to derive the SST kernel from first principles beyond the canonical master equation assumption.

A.2 Canonical constants and reference values

Golden (hyperbolic): $\ln \phi = \operatorname{asinh}(0.5)$, hence $\phi = \varphi$.

(Algebraic form $\phi = (1 + \sqrt{5})/2$ is equivalent.)

The implementation uses the following constants (as literal numeric values in code):

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad (\text{A.1})$$

$$\alpha \equiv \alpha_{\text{fs}} = 7.297\,352\,564\,3 \times 10^{-3}, \quad (\text{A.2})$$

$$c = 299\,792\,458 \text{ m s}^{-1}, \quad (\text{A.3})$$

$$\mathbf{v}_{\text{G}} = 1.093\,845\,63 \times 10^6 \text{ m s}^{-1}, \quad (\text{A.4})$$

$$r_c = 1.408\,970\,17 \times 10^{-15} \text{ m}, \quad (\text{A.5})$$

$$\rho_{\text{core}} = 3.893\,435\,826\,691\,868\,7 \times 10^{18} \text{ kg m}^{-3}, \quad (\text{A.6})$$

$$N_A = 6.022\,140\,76 \times 10^{23} \text{ mol}^{-1}. \quad (\text{A.7})$$

Particle rest masses used for calibration and reporting are taken as fixed numerical targets in the code (electron, muon, tau, proton, neutron).

A.3 Invariant mass kernel

A.3.1 Energy density and effective volume

Define the characteristic kinetic energy density

$$u = \frac{1}{2} \rho_{\text{core}} \mathbf{v}_{\text{G}}^2, \quad (\text{A.8})$$

with units J m^{-3} .

The code uses an effective volume proxy

$$V(T) = \pi r_c^3 L_{\text{tot}}(T), \quad (\text{A.9})$$

where $L_{\text{tot}}(T)$ is dimensionless.

A.3.2 Kernel definition

For each topological object T with invariants $(b_T, g_T, n_T, L_{\text{tot}}(T))$, the invariant mass is

$$M(T) = \left(\frac{4}{\alpha}\right) b_T^{-3/2} \phi^{-g_T} n_T^{-1/\phi} \frac{u V(T)}{c^2}. \quad (\text{A.10})$$

All factors preceding uV/c^2 are dimensionless; therefore the mapping is dimensionally consistent: uV is energy and division by c^2 yields mass.

10-year-old analogy (one sentence). Think of u as how “packed” the motion-energy is in a material, and $V(T)$ as how much of that packed region the string-shape occupies; more packed energy times more occupied volume means more mass.

A.4 Topology-to-geometry assembly and computation paths

A.4.1 Lepton calibration

A base topology for each lepton is specified as:

$$\text{electron base: } (b, g, n) = (2, 1, 1), \quad (\text{A.11})$$

$$\text{muon base: } (b, g, n) = (5, 2, 1), \quad (\text{A.12})$$

$$\text{tau base: } (b, g, n) = (7, 3, 1). \quad (\text{A.13})$$

Given a target mass M_{target} and fixed (b, g, n) , the code solves for the required ropelength by rearranging (A.10):

$$L_{\text{tot}}(T) = \frac{M_{\text{target}} c^2}{\left(\frac{4}{\alpha}\right) b^{-3/2} \phi^{-g} n^{-1/\phi} u \pi r_c^3}. \quad (\text{A.14})$$

This step makes each listed lepton mass exact for its chosen base topology (it is a calibration, not a prediction).

A.4.2 Baryon-sector ropelength mapping

The baryon sector uses fixed $(b, g, n) = (3, 2, 3)$ and constructs baryon ropelengths from two geometric factors (s_u, s_d) via

$$L_{\text{tot}}(p) = \lambda_b (2\pi^2 \kappa_R) (2s_u + s_d), \quad (\text{A.15})$$

$$L_{\text{tot}}(n) = \lambda_b (2\pi^2 \kappa_R) (s_u + 2s_d), \quad (\text{A.16})$$

with $\kappa_R \approx 2$ (default $\kappa_R = 2$ in code).

Modes.

- **canonical:** (s_u, s_d) fixed by constants (hyperbolic-volume assignments external to this script); $\lambda_b = 1$.
- **sector_norm:** (s_u, s_d) fixed as in canonical; λ_b chosen so the proton mass is exact.
- **exact_closure (default):** (s_u, s_d) solved analytically so that *both* proton and neutron are exact with $\lambda_b = 1$.

A.4.3 Analytic closure used in exact_closure

Let $A(b, g, n)$ denote the kernel prefactor multiplying L_{tot} :

$$A(b, g, n) = \left(\frac{4}{\alpha}\right) b^{-3/2} \phi^{-g} n^{-1/\phi} \frac{u \pi r_c^3}{c^2}. \quad (\text{A.17})$$

Define $S \equiv 2\pi^2 \kappa_R$ and $K \equiv A(3, 2, 3) S$. Then

$$M_p = K(2s_u + s_d), \quad (\text{A.18})$$

$$M_n = K(s_u + 2s_d). \quad (\text{A.19})$$

Solving yields

$$s_u = \frac{2M_p - M_n}{3K}, \quad (\text{A.20})$$

$$s_d = \frac{M_p}{K} - 2s_u. \quad (\text{A.21})$$

This is the exact algebra implemented by `fit_quark_geom_factors_for_baryons`.

A.5 From nucleons to atoms and molecules

A.5.1 Atomic mass assembly and binding correction

For a chosen nuclide (Z, N) with $A = Z + N$, the script constructs

$$M_{\text{sum}}(Z, N) = Z M_p^{\text{pred}} + N M_n^{\text{pred}} + Z M_e^{\text{pred}}. \quad (\text{A.22})$$

It then subtracts a semi-empirical mass defect (binding energy divided by c^2),

$$M_{\text{atom}}^{\text{pred}}(Z, N) = M_{\text{sum}}(Z, N) - \Delta m(Z, N), \quad (\text{A.23})$$

where $\Delta m(Z, N)$ is computed via the semi-empirical mass formula (SEMF)

$$B(Z, N) = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(N-Z)^2}{A} + \delta(A, Z) a_p A^{-1/2}, \quad (\text{A.24})$$

$$\Delta m(Z, N) = \frac{B(Z, N)}{c^2}. \quad (\text{A.25})$$

The code uses a MeV→kg conversion factor to implement Δm directly in kilograms. Coefficients are fixed in the script:

$$(a_v, a_s, a_c, a_a, a_p) = (15.75, 17.8, 0.711, 23.7, 11.18) \text{ MeV}.$$

A.5.2 Molecules

Chemical binding energies are neglected (eV scale). A molecular mass prediction is the sum of predicted atomic masses of constituents:

$$M_{\text{mol}}^{\text{pred}}(\text{formula}) = \sum_i \nu_i M_{\text{atom}}^{\text{pred}}(Z_i, N_i), \quad (\text{A.26})$$

where ν_i are stoichiometric coefficients.

A.5.3 Reference conversion from molar mass

For each table entry with molar mass M_{mol} (g/mol), the script uses

$$M_{\text{ref}} = \frac{M_{\text{mol}} \times 10^{-3}}{N_A} \quad [\text{kg}], \quad (\text{A.27})$$

to obtain a per-entity reference mass.

A.6 Results (from the provided run)

This section records the numerical outcomes printed by the script for the run corresponding to *exact_closure* (the observed command-line mode string in the log is non-canonical but triggers the exact-closure branch in the current implementation).

A.6.1 Fitted/derived geometric quantities

With $\kappa_R = 2$,

$$S = 2\pi^2\kappa_R = 39.478\,417\,604\,357\,43, \quad (\text{A.28})$$

$$A(3, 2, 3) = 4.653\,602\,747\,355\,529 \times 10^{-30}, \quad (\text{A.29})$$

$$K = A(3, 2, 3) S = 1.837\,168\,726\,248\,866\,3 \times 10^{-28}. \quad (\text{A.30})$$

Exact-closure geometric factors (dimensionless):

$$s_u = 3.030\,598\,705\,996\,29, \quad (\text{A.31})$$

$$s_d = 3.043\,148\,311\,903\,786\,8, \quad (\text{A.32})$$

$$\lambda_b = 1. \quad (\text{A.33})$$

Calibrated ropelengths (dimensionless):

$$L_{\text{tot}}(e) = 0.033\,396\,273\,374\,363\,69, \quad (\text{A.34})$$

$$L_{\text{tot}}(\mu) = 44.165\,137\,009\,705\,73, \quad (\text{A.35})$$

$$L_{\text{tot}}(\tau) = 1990.712\,148\,304\,272, \quad (\text{A.36})$$

$$L_{\text{tot}}(p) = 359.425\,162\,502\,426\,64, \quad (\text{A.37})$$

$$L_{\text{tot}}(n) = 359.920\,601\,085\,212\,9. \quad (\text{A.38})$$

Reported ratios:

$$\frac{L_\mu}{L_e} \approx 1322.4570, \quad (\text{A.39})$$

$$\frac{L_\tau}{L_\mu} \approx 45.0743. \quad (\text{A.40})$$

A.6.2 Benchmark summary statistics

Define percentage error as

$$\varepsilon = 100\% \times \frac{M^{\text{pred}} - M^{\text{ref}}}{M^{\text{ref}}}.$$

For the elements table (92 entries, H–U) in *exact_closure*:

$$\text{MAE}(|\varepsilon|) \approx 0.196\%, \quad (\text{A.41})$$

$$\text{median}(|\varepsilon|) \approx 0.0690\%, \quad (\text{A.42})$$

$$\text{RMSE}(|\varepsilon|) \approx 0.374\%, \quad (\text{A.43})$$

$$\max |\varepsilon| \approx 1.84\%, \quad (\text{A.44})$$

$$\text{p95}(|\varepsilon|) \approx 0.808\%. \quad (\text{A.45})$$

For the molecules table (19 entries):

$$\text{MAE}(|\varepsilon|) \approx 0.156\%, \quad (\text{A.46})$$

$$\text{median}(|\varepsilon|) \approx 0.0386\%, \quad (\text{A.47})$$

$$\max |\varepsilon| \approx 1.33\%. \quad (\text{A.48})$$

A.6.3 Selected examples

Table A.1 lists representative entries (*exact_closure*).

Table A.1: Selected benchmark entries in *exact_closure*.

Object	Actual mass (kg)	Predicted mass (kg)	ε (%)
H	$1.673\,558 \times 10^{-27}$	$1.673\,533 \times 10^{-27}$	-0.001 48
He	$6.646\,477 \times 10^{-27}$	$6.656\,203 \times 10^{-27}$	0.146
C	$1.994\,473 \times 10^{-26}$	$1.993\,101 \times 10^{-26}$	-0.0688
O	$2.656\,696 \times 10^{-26}$	$2.656\,312 \times 10^{-26}$	-0.0145
Fe	$9.273\,280 \times 10^{-26}$	$9.287\,658 \times 10^{-26}$	0.155
Au	$3.270\,764 \times 10^{-25}$	$3.270\,622 \times 10^{-25}$	-0.004 35
U	$3.952\,581 \times 10^{-25}$	$3.952\,694 \times 10^{-25}$	0.002 85
H ₂ O	$2.991\,461 \times 10^{-26}$	$2.991\,018 \times 10^{-26}$	-0.0148
CO ₂	$7.308\,032 \times 10^{-26}$	$7.305\,725 \times 10^{-26}$	-0.0316
C ₈ H ₁₀ N ₄ O ₂	$3.224\,601 \times 10^{-25}$	$3.223\,412 \times 10^{-25}$	-0.0369

A.6.4 Largest deviations in the fixed table

Table A.2 lists the largest absolute errors over the element and molecule sets.

Table A.2: Largest absolute percentage errors in *exact_closure* for the fixed (Z, N) table used by the script.

Object	Actual mass (kg)	Predicted mass (kg)	ε (%)
B	$1.795\,043 \times 10^{-26}$	$1.828\,057 \times 10^{-26}$	1.839
Cl	$5.886\,611 \times 10^{-26}$	$5.806\,218 \times 10^{-26}$	-1.366
Mg	$4.035\,940 \times 10^{-26}$	$3.982\,563 \times 10^{-26}$	-1.323
Li	$1.152\,414 \times 10^{-26}$	$1.164\,990 \times 10^{-26}$	1.091
Ne	$3.350\,968 \times 10^{-26}$	$3.319\,443 \times 10^{-26}$	-0.941
HCl	$6.054\,325 \times 10^{-26}$	$5.973\,572 \times 10^{-26}$	-1.334
NaCl	$9.704\,190 \times 10^{-26}$	$9.623\,190 \times 10^{-26}$	-0.835

A.7 Interpretation and limitations

(1) Table mismatch: isotopes vs standard atomic weights. The element table fixes a single (Z, N) per element, while the “actual” mass is computed from a standard molar mass (often an isotopic average). A strict test should compare against isotope-specific atomic masses (e.g., AME tables) for the chosen (Z, N) .

(2) Nuclear binding model is semi-empirical. The SEMF step (A.24) is a phenomenological correction. The reported accuracy for atoms therefore reflects the combined effect of (i) exact nucleon calibration in *exact_closure*, and (ii) an empirical nuclear binding proxy.

(3) Leptons beyond the electron are not predicted in this implementation. Muon and tau ropelengths are solved from their measured masses via (A.14); they should be treated as calibrations unless an independent SST rule for lepton L_{tot} is introduced.

(4) Mode-handling robustness. In the current code, any unrecognized `mode` string falls through to the *exact_closure* branch. For publication-quality reproducibility, the CLI

should validate mode strings and fail loudly on invalid values.

A.8 Falsifiable checks enabled by this kernel

Even within this script’s structure, two immediate falsifiers can be stated:

- **Canonical baryons:** Fix (s_u, s_d) from hyperbolic-volume assignments and set $\lambda_b = 1$. The resulting proton/neutron residuals become a direct test of the Canon mapping (independent of the exact-closure fit).
- **Isotope-resolved benchmark:** Replace standard atomic weights with isotope masses for the script’s chosen (Z, N) (or update (Z, N) to match dominant isotopes) and re-evaluate error statistics.

A.9 Reproducibility

The script writes `SST_Invariant_Mass_Results_{mode}.csv` and optionally a cross-mode comparison CSV. For non-interactive environments, the optional prompt should be disabled or guarded by `sys.stdin.isatty()`.

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